

BOUNDARY LAWS AND SMOOTH CONTACT RIGIDITY OF PERIODIC DISPERSING BILLIARDS

QIAN QI

ABSTRACT. Conditional endpoint laws near a supplied nongrazing periodic billiard polygon determine the complete smooth reflecting contact germs, including their flat parts. The success mass is exponentially small. An exact Jacobi normalization and a two-ended determinant comparison produce a relative physical law on a fixed contact collar. Two positive excess times separate its half-line actions without supplied curvature or flux amplitudes. A global positive quadratic inverse, signed finite-jet alignment and the stationary envelope over contracted later visits then identify the actual smooth graphs, without analyticity or a closeness assumption. The same argument gives real-profile stability under a finite-smoothness prior. Actual flat perturbations and equal-area remote completions distinguish this result from formal-jet determination and whole-boundary rigidity. For oblique polygons, a single law per phase also determines its unreported offset; normalized Liouville success counts identify the free area. Moment reconstruction and charged finite-sample estimates are derived consequences of this geometric mechanism.

A phase-weighted envelope estimate replaces a worst-phase sufficient order criterion by one involving the geometric mean of uniform one-step contraction bounds, with an explicit weight-conditioning cost.

CONTENTS

1.	Introduction	2
Part 1.	The relative law and the smooth inverse	5
2.	Relative boundary laws along a periodic itinerary	5
3.	Contact reconstruction along a marked periodic orbit	14
4.	Global identification of the positive quadratic parameters	19
5.	Smooth contact rigidity from the complete boundary law	21
6.	Local determination and exterior scattering data	29
Part 2.	Consequences for the observation record	31
7.	One uncalibrated law at an oblique contact	31
8.	The complete invariant of the charged local experiment	36
9.	Reconstruction from first-moment profiles	42
10.	Conditioning of the physical-to-smooth inverse	48
Appendix A. Smooth contact recovery from finite preparations		50
References		54

Date: September 17, 2026.

2020 Mathematics Subject Classification. 37D50, 37C30, 70H09, 53A04, 62G05.

Key words and phrases. Dispersing billiards, relative boundary laws, smooth contact rigidity, stationary envelope, finite preparations.

1. INTRODUCTION

A long billiard trajectory constrained to follow a hyperbolic periodic orbit has exponentially small phase volume. Its endpoints nevertheless range over fixed neighborhoods of the reflecting contacts. We ask whether the conditional distribution of these endpoints determines the reflecting arcs as smooth functions. Recovering their Taylor series is not sufficient: a nonzero smooth perturbation can be flat at every marked contact.

We prove that two positive excess-time laws at each phase determine the complete smooth contact germs. The polygon and tangent frames are supplied, but the curvatures, graph heights and local flux amplitudes are not. The argument has two substantive steps. First, an exact Jacobi normalization and a two-ended determinant comparison preserve the nonlinear physical law at its exponentially small scale. Second, the stationary envelope on actual reflecting graphs separates the boundary value at the initial contact from all later, contracted visits. A global positive quadratic inverse and a finite signed jet alignment make this second step applicable without analyticity or a closeness assumption.

1.1. The experiment and the main theorem. Fix a clear nongrazing periodic polygon with $r \geq 2$ distinct physical contacts in a planar periodic dispersing billiard. Its lifted vertices q_i , signed unit tangents t_i , outward obstacle normals n_i , phase labels and closing translation are fixed marks. The reference polygon satisfies specular reflection. Write

$$L_i = |q_{i+1} - q_i|, \quad e_i = (q_{i+1} - q_i)/L_i, \quad \nu_i = e_i \cdot n_i > 0, \quad p_i = e_i \cdot t_i/\nu_i.$$

The unknown arcs are the anchored normal graphs

$$R_i(x) = q_i + \nu_i^{-1}\{t_i x - n_i F_i(x)\}, \quad F_i(0) = F'_i(0) = 0, \quad F''_i(0) > 0. \quad (1.1)$$

The observed coordinate $x = \nu_i t_i \cdot (R_i - q_i)$ is a tangent projection. It reveals neither $F_i(x)$ nor an arclength conversion. Clearance means that the open reference flights meet no unintended obstacle; the precise uniform margins are given in Section 2.

Let P be a returning oriented period, doubled when necessary to return the signed frames, and let $\mathcal{L}$ be its length. For $N = nP$ flights starting at phase b , let $W_{N,b}(u, v)$ be the stationary bridge length in the prescribed collars. Set

$$E_{N,b} = W_{N,b} - n\mathcal{L} + p_b u - p_b v, \quad D_{N,b} = -W_{N,b,uv}(0, 0) > 0.$$

At time $n\mathcal{L} + d$, success requires the specified first collision, these N flights in the itinerary collars, no further collision, and endpoints in one fixed small square J^2 . Here $d > 0$ is fixed as N increases. The square is chosen so that the residual time is positive and smaller than either adjacent free-flight length.

Preparations are independent draws from normalized unit-speed Liouville measure on the full phase space. Itinerary selection, gate membership and success/failure tags are exact. The conditional-law theorem does not require the free-area normalizer; its reconstruction from counts below does require this specified preparation measure. Bounded sensor errors in the finite experiment affect accepted tangent-coordinate readouts only.

Theorem 1.1 (Relative boundary laws and smooth contact determination). *For a marked polygon as above with positive-curvature C^∞ contacts, the following assertions hold on a contact collar independent of N .*

There are anchored smooth half-line actions S_b^-, S_b^+ and positive smooth amplitudes B_b^-, B_b^+ , normalized by $B_b^\pm(0) = 1$, such that, for each separately fixed finite order k ,

$$\|E_{N,b} - S_b^- \oplus S_b^+\|_{C^k} + \|(-W_{N,b,uv})/D_{N,b} - B_b^- \otimes B_b^+\|_{C^k} \leq C_k \tau^N, \quad 0 < \tau < 1. \quad (1.2)$$

The reference is exact:

$$D_{nP,b} = \frac{\sinh \chi}{(\mathcal{M}_b)_{12} \sinh(n\chi)}, \quad \text{tr } \mathcal{M}_b = 2 \cosh \chi. \quad (1.3)$$

The monodromy $\mathcal{M}_b$ is computed from the actual geometric Jacobi operator; it is not additional observation data.

For two fixed offsets $0 < d_1 < d_2$ in the positive physical window, the conditional densities on J^2 converge in each fixed C^k norm to

$$f_{b,d}(u, v) = Z_{b,d}^{-1} B_b^-(u) B_b^+(v) \{d - A_b(u) - C_b(v)\}, \quad d \in \{d_1, d_2\}, \quad (1.4)$$

where $A_b = S_b^- - p_b \text{id}$ and $C_b = S_b^+ + p_b \text{id}$. The positive constants $Z_{b,d}$ are not supplied. Equality of these phase-resolved laws for two actual contact tuples implies equality of their smooth contact germs in (1.1). No curvature input, reflection symmetry, analyticity or closeness between the candidates is required.

On a fixed bounded positive class, an integer m and a common smaller collar I_c can be chosen so that the inverse also satisfies

$$\|F - G\|_{C^0(I_c)} \leq C \max_{b,d} \|f_{b,d}^F - f_{b,d}^G\|_{C^m(I^2)} \quad (1.5)$$

for sufficiently small right side, with $I_c \Subset I \subset J$. The class has the C^{m+3} graph bounds and positive quadratic bounds of (5.7); the densities retain their normalization on J^2 when restricted to I^2 . The order m is chosen from the contraction margin, not uniformly over degenerating geometric classes.

Proof. The relative and phase-volume statements are Theorems 2.3 and 2.5, transferred to measured graph coordinates by Lemma 3.1. Proposition 3.2 recovers the actions. Theorem 4.2 identifies their positive quadratic parameters globally, and Theorem 3.4 aligns finite jets. The actual-function estimate in Proposition 5.2 proves smooth uniqueness and, after polynomial alignment, Theorem 5.5 proves (1.5). All these proofs are included below. $\square$

1.2. The two analytic steps. An absolute estimate for the stationary action does not control a conditional law whose mass is of order $D_{N,b}$. The exact unconditioned density, with free cell area $\mathcal{A}$, is

$$\frac{D_{N,b}}{2\pi\mathcal{A}} \frac{-W_{N,b,uv}(u, v)}{D_{N,b}} \{d - E_{N,b}(u, v) + p_b u - p_b v\}. \quad (1.6)$$

Thus the required estimate is relative before conditioning, and the linear endpoint momenta must be restored before the clock is read. The cofactor formula cancels $D_{N,b}$ exactly. Its logarithm contains a sum of local mixed-chord terms and $\log \det(I + G_N \Delta H_N)$, where $G_N = H_N(0)^{-1}$. Although the number of interior sites increases, endpoint localization keeps ΔH_N uniformly trace class. Separating two end blocks, controlling the opposite-end Green entries and differentiating the convergent logarithmic series prove (1.2). A quadratic approximation, or an absolute action error divided by the success probability, would not provide this estimate.

The action extraction is elementary once this physical limit is known. At one phase set

$$Q(u, v) = \frac{f_{d_1}(u, v) f_{d_2}(0, 0)}{f_{d_2}(u, v) f_{d_1}(0, 0)}.$$

Then

$$\frac{d_1 d_2 (1 - Q)}{d_2 - d_1 Q} = A(u) + C(v).$$

The denominator has a positive floor on the common gate. Axis evaluation recovers the two actions without knowing either flux amplitude.

The passage from actions to graphs is functional, not formal. The quadratic Schur complement gives a global inverse on the positive curvature image. With these curvatures fixed, the signed cyclic response at each higher order is invertible. This identifies the finite jets required by the next argument, but it cannot itself exclude a flat boundary difference. For two actual graph tuples, integrate the stationary envelope along their local interpolation to obtain

$$\mathcal{S}(G) - \mathcal{S}(F) = \bar{\alpha}(G - F) + K(G - F), \quad \bar{\alpha} \geq \frac{1}{2}. \quad (1.7)$$

Every summand of K evaluates a boundary difference at a later visit $x_{b,j}(u)$ satisfying $|x_{b,j}(u)| \leq a^j|u|$ with $a < 1$ and multiplicative constant one. For differences vanishing through order $m - 1$,

$$\|h\|_{m,R} = \max_b \sup_{0 < |u| \leq R} \frac{|h_b(u)|}{|u|^m}, \quad \|\bar{\alpha}^{-1}K\| \leq \frac{6a^m}{1 - a^m} < 1$$

after choosing a finite m . This separates actual smooth functions, including their flat parts. The terminal term of the finite stationary envelope is estimated before the half-line limit is taken. Polynomial jet alignment makes the same estimate quantitative for unequal data.

Proposition 5.6 realizes the distinction on closed obstacles: an $\varepsilon\chi(u)e^{-1/u^2}$ contact perturbation, compensated by a remote area adjustment, preserves every contact and action Taylor jet but changes the complete boundary laws. The local response is not manufactured by a change of normalization. The complementary locality result in Section 6 identifies what the observation leaves free away from the visited arcs.

The two estimates admit a formulation independent of the billiard notation. Proposition 2.7 isolates the relative comparison of two localized end blocks: the first off-diagonal variation of the logarithmic determinant vanishes, so their coupling enters quadratically in Hilbert–Schmidt norm, with no dimension factor. Proposition 5.7 bounds an actual integrated envelope by a nonnegative visit matrix and inverts it when its spectral radius is less than one. These statements express the analytic mechanism used here; they do not identify different observation models.

For periodic visits the latter estimate has a quantitative refinement. If $\hat{a}_i < 1$ are common one-step bounds on the interpolation class, set $\bar{a} = (\prod_i \hat{a}_i)^{1/r}$. Corollaries 5.8 and 5.9 replace the sufficient scalar condition $6a^m/(1 - a^m) < 1$ by $6\bar{a}^m/(1 - \bar{a}^m) < 1$. Their explicit phase weights carry a condition-number cost in the unweighted estimate. This improves the sufficient order criterion on heterogeneous classes, not the target or the information supplied to the inverse. It need not improve every conditioning constant.

1.3. What the observation refinements add. The remaining local results use this same inverse rather than supplying additional rigidity mechanisms. At oblique phases, the marked nonzero slope identifies an unreported positive offset from one law; the fixed-slice extraction in Section 7 avoids an additional derivative at each inverse order. For normalized Liouville preparations, the same-gate success count identifies the free-area scalar. Section 8 proves both directions of the resulting invariant: contact germs, offsets and free area. The exact area identity uses the actual finite density, and its stability retains the factor $1 + N$ from logarithmic twist sensitivity. It does not determine unvisited boundary arcs.

Section 9 recovers the same limiting density from four one-dimensional weighted profiles through $MH^{-1}N$. The cancellation is the classical low-rank mechanism, as in [7]; the application requires signed geometric nondegeneracy and the finite-flight defect. The profiles retain paired endpoint information. They are not exact sufficient statistics for an arbitrary finite-flight density. On a fixed oblique gate they reduce the smoothing dimension, giving the sufficient exponents

$$\alpha_1 = \frac{s}{2(m+s)+1}, \quad \gamma_1 = \frac{s}{m+s+2}, \quad \beta_1 = \frac{\alpha_1\omega}{\omega + \alpha_1\Gamma}.$$

With the explicit count, bandwidth and small-error conditions of Theorem 9.7 and Corollary 9.8, rB charged preparations give profile-and-offset error Ct and joint logarithmic-area error $Ct \log(e/t)$ with probability at least $1 - 2\zeta$, where $t = (L_B/B)^{\beta_1} + \delta^{\gamma_1}$. These are upper bounds for a measurable actual-table estimator, not claims of minimax optimality or effective enumeration. The affine example in Remark 9.3 exhibits the loss of moment conditioning as the gate or obliquity shrinks. Section 10 combines this with the inverse order, phase weights, offset separation, rare acceptance and finite-area sensitivity. These sufficient estimates are not efficiency guarantees.

1.4. Relation to other inverse problems. The corrected planar theorem of Noakes–Stoyanov [5] globally determines finite unions of strictly convex C^3 obstacles from exterior travelling-time or scattering-length data. Smooth obstacle determination from rich exterior data is therefore not new in itself. Here the table is periodic, an internal polygon and its frames are supplied, and the observations are localized conditional endpoint laws. Our smooth target is the visited contact germs. Equal-area remote completions in Corollary 6.3 have the same local record; their exterior scattering data on a common compact realization differ. This explains the spatial scope without identifying the two data sets.

De Simoi–Kaloshin–Leguil [1] use marked lengths for analytic open billiards under symmetry and genericity assumptions. Finamore–Leguil [2] use an enriched marked length spectrum for finite-horizon Sinai billiards. Our result neither removes hypotheses from those problems nor establishes same-data dominance. Its content is the passage from a relatively controlled rare physical law to an inverse sensitive to actual flat boundary changes. The matrix identity, quotient extraction, concentration and minimum-distance selection are supporting tools; they are not offered as new general principles of rigidity or statistics.

1.5. Organization. Sections 2–5 contain the complete central proof: the physical relative law, measured coordinates and stationary envelope, global curvature identification, and smooth uniqueness with stability. Section 6 proves exact locality and constructs actual remote completions. Sections 7–9 develop the oblique single-law protocol, its area-augmented invariant, and the moment reconstruction with finite-rank defect and charged one-dimensional estimation. Section 10 gives the integrated conditioning and budget analysis. Appendix A provides the two-offset finite-preparation argument; it is also needed when a marked phase has normal incidence.

The accompanying *Full technical manuscript* contains the complete analytic, finite-stopping, alternating-channel, intrinsic multichannel, lattice, calibration and coarsening arguments. References prefixed F point to its numbered statements. In the principal article these references identify extensions or special-case comparisons, not omitted premises of the central smooth theorem. In particular, neither analytic continuation nor a global matching hypothesis is used in that proof. The proof organization follows this dependency distinction rather than the order in which observation refinements were obtained.

Part 1. The relative law and the smooth inverse

2. RELATIVE BOUNDARY LAWS ALONG A PERIODIC ITINERARY

The relative normalization is not specific to a normal two-periodic orbit. We prove it for a selected nongrazing periodic orbit of a smooth planar dispersing billiard. The two ends now carry different half-line actions. There is also a linear endpoint contribution to the physical length; it must be restored before a probability law is formed. The subsequent principal argument treats all marked collision phases. Alternating two-contact specializations and the analytic and lattice extensions are retained separately in the full technical manuscript.

2.1. The geometric Jacobi operator. Let a periodic orbit have a fixed itinerary, with lifted reflection points q_i , flight lengths $L_i > 0$, curvatures $\kappa_i > 0$, and outgoing normal cosines $\nu_i > 0$. The indices are periodic modulo an even integer P : an odd primitive period is doubled so that the signed coordinate frames also return. A lifted period is allowed to end at a lattice translate of its initial point. Put $L = \sum_{i=0}^{P-1} L_i$. Every open flight is clear of unintended obstacles. We work with a compact smooth family of such marked orbits, with fixed P and positive lower bounds for clearance, L_i , κ_i and ν_i , and finite upper bounds for the flight lengths and each separately used smooth norm. Clearance here refers to portions outside small fixed endpoint neighborhoods; nongrazing incidence controls those neighborhoods. Family derivatives keep the scaled endpoint coordinates and the excess $d = t - nL$ fixed; the physical observation time moves with the reference period length.

Choose unit-speed signed boundary charts $r_i(s)$ centered at q_i . Their tangent orientations alternate relative to the positively oriented obstacle tangents. Write

$$x_i = \nu_i s_i, \quad \ell_i(x, y) = |r_{i+1}(y/\nu_{i+1}) - r_i(x/\nu_i)|, \quad k_i = L_i^{-1}, \quad m_i = 2\kappa_i/\nu_i.$$

The endpoint arguments in r_i are signed arclengths. Reflection gives numbers p_i such that

$$\ell_{i,1}(0, 0) = -p_i, \quad \ell_{i-1,2}(0, 0) = p_i.$$

Introduce the normalized edge action

$$h_i(x, y) = \ell_i(x, y) - L_i + p_i x - p_{i+1} y. \quad (2.1)$$

Thus $h_i(0, 0) = 0$ and $Dh_i(0, 0) = 0$. The linear terms telescope; this normalization does not change an interior reflection equation or a mixed endpoint derivative.

Lemma 2.1 (Coercivity and persistence). *The Hessian of an edge at the reference orbit and the interior Jacobi operator are respectively*

$$D^2 h_i(0, 0) = \begin{pmatrix} k_i + m_i/2 & -k_i \\ -k_i & k_i + m_{i+1}/2 \end{pmatrix}, \quad (H^0 x)_i = (k_{i-1} + k_i + m_i)x_i - k_{i-1}x_{i-1} - k_i x_{i+1}. \quad (2.2)$$

They are uniformly positive. On a sufficiently small common coordinate box the interior Hessians are symmetric, tridiagonal, and strictly diagonally dominant with a uniform positive margin. The marked orbit and these properties persist under sufficiently small smooth perturbations of the reflecting arcs and of the lattice realization, with the lifted itinerary fixed.

Proof. For a chord with unit direction e and endpoint tangent t , the same-end second derivative in signed arclength is $(1 - (e \cdot t)^2)/L$ plus the curvature contribution. At an outgoing reflection $e \cdot n = \nu$, with n the outward obstacle normal and $r'' = -\kappa n$, that contribution is $\kappa\nu$; the incoming end has the same sign. The mixed derivative is $-[t_i \cdot t_{i+1} - (e \cdot t_i)(e \cdot t_{i+1})]/L$. The alternating tangent orientations make the two projections onto $e^\perp$ have the same sign, so it is $-\nu_i \nu_{i+1}/L_i$. Dividing the endpoint derivatives by the respective normal cosines proves the edge formula. Summing adjacent edge derivatives proves the second formula.

The edge quadratic form is $k_i(x - y)^2 + (m_i/2)x^2 + (m_{i+1}/2)y^2$. The finite and half-line interior forms are therefore bounded below by $m_* \sum x_i^2$, where $m_* := \min_i m_i > 0$. Continuity, finite periodicity, and compactness preserve a positive row-dominance margin on one smaller box. For persistence, apply the finite-dimensional implicit function theorem to the reflection equations with the prescribed periodic identification of the endpoints. The periodic Hessian has quadratic form $\sum k_i(x_{i+1} - x_i)^2 + \sum m_i x_i^2$, hence is invertible. The continued orbit remains in its charts; the clearance and incidence margins keep the chords physical. This argument also covers a repeated period and the prescribed deck translation. $\square$

2.2. The exact reference twist. Set

$$Q_i = \begin{pmatrix} 1 + m_i/k_i & 1/k_i \\ m_i & 1 \end{pmatrix}, \quad \mathcal{M}_b = Q_{b+P-1} \cdots Q_b, \quad \text{tr } \mathcal{M}_b = 2 \cosh \chi. \quad (2.3)$$

Each matrix Q_i has determinant one and positive entries. The products for different starting phases are conjugate. Since $(\mathcal{M}_b)_{12}(\mathcal{M}_b)_{21} > 0$ and $\det \mathcal{M}_b = 1$, their trace is greater than two. In particular $\chi > 0$, and χ has a positive lower bound on the compact class.

For $N = nP$, let $W_{N,b}(u, v)$ be the stationary length along N flights, starting at phase b , with scaled endpoint coordinates u, v . Its construction on a fixed box is given below. Define

$$D_{N,b} = -W_{N,b,uv}(0, 0) = \frac{\sinh \chi}{(\mathcal{M}_b)_{12} \sinh(n\chi)}. \quad (2.4)$$

In particular $D_{N,b} > 0$, and its decay is comparable to $e^{-n\chi}$. This formula is exact, rather than an unspecified exponential equivalent.

Lemma 2.2 (Transfer and cofactor normalizations). *Formula (2.4) holds. More generally, if y is the stationary segment and $H_N(y)$ its interior Hessian, then*

$$-W_{N,b,uv}(u, v) = \frac{\prod_{i=0}^{N-1} [-\ell_{b+i,12}(y_i, y_{i+1})]}{\det H_N(y)}. \quad (2.5)$$

Proof. For the reference recurrence write $z_i = k_{i-1}(x_i - x_{i-1})$. Then $z_{i+1} = z_i + m_i x_i$ and $x_{i+1} = x_i + z_{i+1}/k_i$, which is precisely the transfer Q_i . At the initial endpoint z_0 can be defined as $k_b(x_1 - u) - m_b u$; no exterior reflection equation is being imposed. The left boundary derivative of the quadratic stationary action is $-z_0 - m_b u/2$. The relation $v = (\mathcal{M}_b^n)_{11}u + (\mathcal{M}_b^n)_{12}z_0$ therefore gives $-W_{uv}(0, 0) = 1/(\mathcal{M}_b^n)_{12}$. Cayley–Hamilton, or diagonalization at the distinct eigenvalues e^χ and $e^{-\chi}$, gives

$$(\mathcal{M}_b^n)_{12} = (\mathcal{M}_b)_{12} \frac{\sinh(n\chi)}{\sinh \chi}.$$

The first assertion follows. Eliminating the interior variables from the tridiagonal Hessian gives the mixed Schur-complement entry. The corner cofactor is the product of the adjacent mixed entries, with the sign in (2.5). This proves the second assertion, including its normalization at zero. $\square$

2.3. A relative theorem on a fixed collar. Write

$$E_{N,b}(u, v) = W_{N,b}(u, v) - nL + p_b u - p_b v, \quad \beta_{N,b}(u, v) = \frac{-W_{N,b,uv}(u, v)}{D_{N,b}}.$$

The superscripts $-$ and $+$ below distinguish the left future tail from the right past tail. They do not indicate signs of the functions.

Theorem 2.3 (Periodic-itinerary relative boundary law). *For the marked periodic orbits above there is a coordinate interval, independent of n , on which the finite stationary segments exist uniquely within the prescribed small itinerary collars. There are smooth strictly convex functions S_b^-, S_b^+ and positive smooth functions B_b^-, B_b^+ , with*

$$S_b^\pm(0) = (S_b^\pm)'(0) = 0, \quad B_b^\pm(0) = 1,$$

such that, for one $\tau \in (0, 1)$ and each fixed k ,

$$\|E_{N,b} - S_b^- \oplus S_b^+\|_{C^k} + \|\beta_{N,b} - B_b^- \otimes B_b^+\|_{C^k} \leq C_k \tau^N, \quad N = nP. \quad (2.6)$$

The norms include each stipulated fixed number of smooth family derivatives. Constants may depend on the corresponding higher norms; no order-uniform smooth estimate is asserted. The action error has zero value and endpoint differential at zero and is $O(\tau^N(|u| + |v|)^2)$. The

amplitude error vanishes at zero. For analytic families all four limits are analytic on smaller common complex collars.

Proof. We give the Green estimates, the construction, and the relative normalization explicitly. None of them uses a two-dimensional normalizing chart.

Put $a_i = k_{i-1} + k_i + m_i$, and split the reference interior operator as $H^0 = A(I - T)$, where $A = \text{diag}(a_i)$ and the two nonzero entries in row i of T are k_{i-1}/a_i and k_i/a_i . On a finite interval or a half-line,

$$\|T\|_\infty \leq q < 1, \quad (H^0)^{-1} = \sum_{r \geq 0} T^r A^{-1}, \quad |G_{ij}| \leq Cq^{|i-j|}.$$

The last estimate holds because a nearest-neighbor walk from i to j has length at least $|i - j|$. Symmetry also gives uniform ℓ^1 bounds. The same series is bounded on the exponentially weighted sup norm for any weight $\rho \in (q, 1)$, by summing $q^{|i-j|}\rho^{j-i}$. For a finite segment one uses the sum of the two endpoint weights $\rho^i + \rho^{N-i}$.

These estimates include fixed derivatives in the marked family. A derivative of order r of a product representing a walk of length h introduces at most a fixed polynomial in h , times a bound $C_r q^{h-r}$. Alternatively one may differentiate the resolvent identity. A strict enlargement of the exponential base absorbs that polynomial. The finite Green kernel and the left half-line kernel also satisfy

$$|G_{ij}^N - G_{ij}^-| \leq Cq^{2N-i-j} \quad (i, j < N), \quad |G_{ij}^N| \leq Cq^{|i-j|}. \quad (2.7)$$

Indeed a walk contributing to their difference must reach the removed right boundary and return. Its length is at least $2N - i - j$. The reflected right version and the fixed differentiated versions follow in the same way, with a slower base when necessary. The diagonal coefficients on retained sites are identical; imposing a Dirichlet boundary deletes walks, not those diagonal coefficients.

Solve the finite reflection equations as their reference linear solution minus G^N times the nonlinear gradient remainder. That remainder is local, vanishes quadratically at zero, and has derivative $O(r)$ in a box of radius r . The weighted Green bounds make this a contraction with constant at most $1/2$ on a ball of radius $C(|u| + |v|)$ when the common endpoint interval is sufficiently small. This constructs the physical segment with

$$|y_i| \leq C(|u|\rho^i + |v|\rho^{N-i}).$$

Strict convexity gives uniqueness among all segments in the small box, not just in the contraction ball. The same construction on the future and past half-lines gives $x_i^-(u)$ and $x_i^+(v)$, decaying as $C|u|\rho^i$ and $C|v|\rho^i$. Differentiating the fixed-point equation leaves the same invertible operator on its highest derivative; the other terms are local products of lower derivatives. Induction therefore gives the weighted bounds at every fixed order.

Define the half-line actions by the absolutely convergent sums

$$\begin{aligned} S_b^-(u) &= \sum_{i \geq 0} h_{b+i}(x_i^-(u), x_{i+1}^-(u)), \\ S_b^+(v) &= \sum_{i \geq 0} h_{b-i-1}(x_{i+1}^+(v), x_i^+(v)). \end{aligned} \quad (2.8)$$

Every summand is quadratic in its exponentially decaying arguments. The fixed differentiated sums converge as well. In the first variation of a finite truncation the interior terms cancel. The remaining remote boundary term tends to zero by the weighted estimates, so these sums have the stated endpoint derivatives. The positive edge forms in Lemma 2.1 bound each minimized half-line energy below by a fixed positive multiple of its boundary value squared. Their Hessians

at zero are consequently positive. Shrinking the interval proves strict convexity and the stated anchoring.

For the action comparison, glue the two half-lines by $z_i = x_i^-(u) + x_{N-i}^+(v)$ and overwrite the two endpoint values by u, v . Locality and $Dh_i(0, 0) = 0$ imply that the interior residual has ℓ^1 norm at most $C(1 + N)\rho^N$: every interaction contains one factor from each end. The endpoint overwrites have the same order. The averaged Hessian between z and the finite stationary segment has uniformly bounded ℓ^1 inverse by diagonal dominance and symmetry. Hence $\|y - z\|_1 \leq C(1 + N)\rho^N$. Subtracting and differentiating these equations proves the same assertion at each fixed order, with only a polynomial loss in N . Summing edge interactions and the omitted half-line tails now proves the action part of (2.6).

For the amplitude, let $G_N = (H_N(0))^{-1}$ and $\Delta H_N = H_N(y) - H_N(0)$. The exact cofactor identity gives

$$\log \beta_{N,b} = \sum_{i=0}^{N-1} \log \frac{-\ell_{b+i,12}(y_i, y_{i+1})}{k_{b+i}} - \log \det(I + G_N \Delta H_N). \quad (2.9)$$

In particular the exponentially small $D_{N,b}$ has already been cancelled exactly. Tridiagonality and endpoint localization give $\|\Delta H_N\|_1 \leq C(|u| + |v|)$, where the norm is the trace norm: the sum of absolute matrix entries is an upper bound for that norm. All its fixed derivatives have uniformly bounded trace norms. Take the endpoint interval small enough that $\|G_N \Delta H_N\| < 1/2$, uniformly in N .

On the future half-line define $\Delta H_b^-(u)$ by the same rule, and put

$$\begin{aligned} \log B_b^-(u) &= \sum_{i \geq 0} \log \frac{-\ell_{b+i,12}(x_i^-(u), x_{i+1}^-(u))}{k_{b+i}} \\ &\quad - \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} \operatorname{tr}[(G_b^- \Delta H_b^-(u))^r]. \end{aligned} \quad (2.10)$$

The analogous reversed formula defines B_b^+ . Each edge sum is absolutely convergent. The perturbation is trace class and the operator norm is less than $1/2$, so the trace series and its fixed derivatives converge. Its exponential is positive for real arguments, and equals one at zero. This specifies the determinant normalization completely.

Here is the two-end comparison for (2.9). For $N \geq 6$ retain the first and last $r = \lfloor N/3 \rfloor$ interior sites, and delete all other perturbation entries. The trace-norm cost is $C_k(1 + N)^{a_k} \rho^r$ after k fixed derivatives. On either retained block the gluing estimate replaces the finite segment by its own half-line with exponentially small entrywise sum. The identity $\det(I + AB) = \det(I + BA)$ reduces a finitely supported perturbation to those two blocks. Equation (2.7) compares the diagonal Green compressions to their half-line compressions; the opposite-end Green blocks are $O(q^{N-2r})$, up to a polynomial factor in N . Multiplication by the bounded trace-norm perturbation preserves these bounds. Extending each retained block to its half-line costs $O(\rho^r)$ again.

For trace-class $T, \tilde{T}$ with norms at most $q_1 < 1$, the exact telescoping expansion of their powers gives

$$|\operatorname{tr}(T^h - \tilde{T}^h)| \leq h q_1^{h-1} \|T - \tilde{T}\|_1.$$

Summing with coefficient $1/h$ proves the required continuity of the logarithmic determinant. At each fixed differentiated order the series has an additional fixed polynomial in h , still summable. Thus the finite logarithm tends relatively to the sum of the two logarithms in (2.10). The edge-product comparison follows from the same gluing and tail bounds. Exponentiation proves the amplitude part of (2.6).

Choose one $\tau < 1$ slower than the Green, gluing, and block-truncation rates. The strict margins absorb every separately fixed polynomial loss; the constants, not this choice of τ , depend on the order. Finitely many shorter segments are absorbed by enlarging those constants. At zero the actions have zero value and differential, and the normalized twists and amplitudes equal one. Taylor's integral remainder gives the additional quadratic vanishing. Finally the contraction, summable perturbations, and logarithmic series extend to uniform smaller complex boxes for analytic families. Their locally uniform convergence proves the analytic assertion. $\square$

2.4. Nonlinear transmission and the stable return. Define the connected endpoint action and the two increasing coordinates by

$$C_{N,b}(u, v) = W_{N,b}(u, v) - W_{N,b}(u, 0) - W_{N,b}(0, v) + W_{N,b}(0, 0),$$

$$\zeta_b^\pm(x) = \int_0^x B_b^\pm(t) dt.$$

Corollary 2.4 (Relative nonlinear transmission). *On a smaller fixed interval,*

$$-\frac{C_{N,b}(u, v)}{D_{N,b}} = \zeta_b^-(u)\zeta_b^+(v) + R_{N,b}(u, v), \quad \|R_{N,b}\|_{C^k} \leq C_k \tau^N, \quad (2.11)$$

with $|R_{N,b}(u, v)| \leq C\tau^N|uv|$. Let T_b^- be the next-period position on the future stable half-line, and T_b^+ the previous-period position on the past half-line. Then

$$\zeta_b^\pm(T_b^\pm(x)) = e^{-\chi} \zeta_b^\pm(x). \quad (2.12)$$

Each $\zeta_b^\pm$ is the unique differentiable local linearizing coordinate with value zero and derivative one at zero.

Proof. Twice integrating the mixed derivative, with oriented integrals on negative intervals, gives the exact identity

$$-C_{N,b}(u, v)/D_{N,b} = \int_0^u \int_0^v \beta_{N,b}(s, t) dt ds.$$

The relative theorem proves (2.11), including its endpoint vanishing and fixed mixed derivatives. Notice that the unnormalized remainder has the smaller size $O(D_{N,b}\tau^N|uv|)$; an absolute action estimate alone does not give this transmission coefficient.

For the future return concatenate a P -flight segment with an N -flight tail. The joining endpoint $w_N(u, v)$ is uniquely determined by stationarity and tends to $T_b^-(u)$ with every fixed derivative. The denominator in its implicit equation is the sum of two positive same-end Hessians. Exact stationary differentiation gives

$$-W_{N+P,b,uv}(u, v) = [-W_{N,b,12}(w_N(u, v), v)] \partial_u w_N(u, v).$$

Since $D_{N+P,b}/D_{N,b} \rightarrow e^{-\chi}$, relative factorization and cancellation of the positive right factor yield

$$B_b^-(T_b^-(u))(T_b^-)'(u) = e^{-\chi} B_b^-(u).$$

Integration gives the minus identity in (2.12). Reversing the itinerary proves the plus identity. In particular both return maps have derivative $e^{-\chi}$ at zero and are contractions on a smaller interval. If $\tilde{\zeta}$ is another normalized differentiable linearizer, then $h = \tilde{\zeta} \circ (\zeta_b^\pm)^{-1}$ satisfies $h(\lambda x) = \lambda h(x)$, $h(0) = 0$, $h'(0) = 1$, where $\lambda = e^{-\chi}$. Thus $h(x) = \lambda^{-n} h(\lambda^n x) \rightarrow x$, proving uniqueness. $\square$

2.5. The physical window law. The gauge in (2.1) is useful analytically, but it is not a change of the clock. Define the physical boundary actions

$$\Sigma_b^-(u) = S_b^-(u) - p_b u, \quad \Sigma_b^+(v) = S_b^+(v) + p_b v. \quad (2.13)$$

In general they do not have critical points at zero. Their linear parts are the oblique endpoint momenta of the reference orbit.

Theorem 2.5 (Charged phase-volume law for a repeated itinerary). *Let A be the free area of the periodic billiard cell and normalize its unit-speed phase volume by $2\pi A$. Fix a sufficiently small positive offset interval $[d_-, d_+]$. There is a smaller closed endpoint interval I , independent of $N = nP$, such that the following experiment is physical. The first collision is in the phase- b collar; there are the selected N successive flights and no further collision by time $t_N = nL + d$; the two recorded endpoints lie in I^2 . The intervening collisions are required to remain in the prescribed itinerary collars.*

Its endpoint measure, before conditioning on success, has density

$$\frac{D_{N,b}}{2\pi A} \beta_{N,b}(u, v) \{d - E_{N,b}(u, v) + p_b u - p_b v\}_+, \quad (u, v) \in I^2. \quad (2.14)$$

On this interval the expression inside braces is bounded below by $d_-/2$ and above by a constant smaller than either adjacent free-flight length. Write $Z_{N,b,d}$ for the integral of the numerator after the constant $D_{N,b}/(2\pi A)$ is removed. The success probability is exactly $D_{N,b}Z_{N,b,d}/(2\pi A)$ and is bounded above and below by positive constant multiples of $e^{-n\chi}$.

The conditional endpoint density converges, in every fixed $C^k(I^2)$ norm including the stated family derivatives, to

$$f_{b,d}(u, v) = Z_{b,d}^{-1} B_b^-(u) B_b^+(v) \{d - \Sigma_b^-(u) - \Sigma_b^+(v)\}, \quad \|f_{N,b,d} - f_{b,d}\|_{C^k} \leq C_k \tau^N. \quad (2.15)$$

All rejected independent phase preparations remain charged through the success probability. The result is a law for this selected windowed itinerary, not an identification with a maximal-collision event.

Proof. Clearance and nongrazing incidence imply that the actual preceding and following free flights of every sufficiently small stationary segment have a common positive lower bound l_* . Choose $d_+ < l_*/4$. The action has a uniform quadratic bound on the fixed box, whereas $|p_b|$ is uniformly bounded. Shrink I until $|E_{N,b} - p_b u + p_b v| \leq d_-/2$, uniformly in N . Then the available residual time $r = t_N - W_{N,b} = d - E_{N,b} + p_b u - p_b v$ lies in $[d_-/2, 3d_+/2]$, below l_* . Any split $r = s + (r - s)$, $0 \leq s \leq r$, lies strictly within the adjacent flights (apart from measure-zero collision endpoints). It gives exactly the required first and last collision times. Conversely a successful trajectory has this unique first collision, its unique small stationary bridge, and such a split. Thus no multiplicity is introduced by repeating the word.

If x is scaled boundary arclength, its canonical tangential momentum is the ordinary arclength momentum divided by ν_b . Hence $dx \wedge d\pi$ is the usual billiard flux form. The endpoint generating action gives $\pi = -W_{N,b,u}$, so that the absolute Jacobian to (u, v) is $-W_{N,b,uv} > 0$. The full phase form is this flux times ds . Integrating over $0 \leq s \leq r$ proves (2.14), with precisely the normalization $2\pi A$. In particular the endpoint gauge terms have been restored.

The residual interval has a positive floor. The finite and limiting amplitudes have positive upper and lower bounds on a common box, by (2.9) and (2.10). Consequently $Z_{N,b,d}$ and $Z_{b,d}$ have positive uniform floors and finite uniform upper bounds. The relative theorem compares their unnormalized integrands and fixed derivatives at rate $C_k \tau^N$. Integration on the fixed square and division by these floors prove (2.15). Finally (2.4) gives the stated charged probability. At no stage is an absolute action error divided by that probability. $\square$

2.6. A nonnormal orbit and the alternating specialization.

Proposition 2.6 (A three-obstacle realization). *There is a periodic dispersing table with a nongrazing three-periodic orbit to which Theorems 2.3 and 2.5 apply, and an open family of such realizations with unequal flight lengths and incidence angles.*

Proof. In a square cell of side 12 place three unit disks whose centers form an equilateral triangle of side $D = 5/2$, centered in the cell. The disks and their translates are disjoint. At each disk choose the point on its radius directed toward the triangle center. These three points form an equilateral triangle of side $\ell = D - \sqrt{3} > 0$. Its inscribed three-flight path satisfies reflection: the outward disk normal bisects the two outgoing directions, with normal cosine $\nu = \sqrt{3}/2$. The distance from the opposite disk center to a whole side line is $\sqrt{3}D/2 - 1/2 > 1$; hence no third disk blocks that side. Neighboring cells are much farther away. The flights are therefore clear and the incidence is not normal or grazing.

Take the oriented period $P = 6$. Then $k_i = 1/\ell$, $m_i = 2/\nu$, and

$$\chi = 6 \operatorname{arccosh}(1 + \ell/\nu).$$

For explicit nonsymmetric realizations, choose radii R_0, R_1, R_2 close to one and prescribe a nearby scalene contact triangle. Place center i at distance R_i behind vertex i along its interior-angle bisector. The triangle then satisfies reflection by construction, has unequal sides and, choosing its three angles distinct, unequal normal cosines. Disjointness and clearance persist by the strict margins just checked. This gives the asserted open family by the same persistence lemma. $\square$

For a normal alternating channel take $P = 2$, $L_0 = L_1 = g$ and $\nu_0 = \nu_1 = 1$. Then $p_b = 0$, $\chi = 2\gamma$, and (2.4) reduces to $D_{2n,b} = a_b / \sinh(2n\gamma)$. Reversal identifies the two half-line actions and amplitudes. Passing from arclength to the fixed transverse graph chart changes an amplitude by the corresponding endpoint Jacobian, normalized by its value at zero. Thus the relative law specializes to Theorem F.17.2, not to a different normalization of it. The intrinsic scalar coordinate in Corollary 2.4 specializes to Theorem F.19.3.

The extension above concerns the geometric forward mechanism and its physical normalization. It neither assumes nor proves that the two boundary actions of a general itinerary can be inverted by the alternating two-contact block formula. The next sections establish the general periodic finite-jet and smooth inverses in measured graph coordinates. The alternating inverse, analytic extensions and global lattice reconstruction in the full technical manuscript retain their separate hypotheses. The finite-preparation consequences of the smooth inverse are proved in the principal article and its two-offset appendix.

2.7. A dimension-independent two-end comparison. The determinant argument admits a formulation that separates the size of an experiment from the interaction of its two ends. We state it for finite matrices of arbitrary size; padding a matrix by zero does not change any of the quantities below. The trace and Hilbert–Schmidt norms are denoted by $\|\cdot\|_1$ and $\|\cdot\|_2$, respectively. For a parameter domain U , a $C^k(U; \mathfrak{S}_p)$ bound means the maximum of these norms over all parameter derivatives of order at most k . Operator norms will be written without a subscript.

Proposition 2.7 (Separated perturbation determinants). *Fix $k \geq 0$ and $q < 1$. Relative to two orthogonal end spaces, write*

$$T_N = D_N + E_N, \quad D_N = \operatorname{diag}(T_{N,-}, T_{N,+}), \quad E_N = \begin{pmatrix} 0 & E_{N,-+} \\ E_{N,+-} & 0 \end{pmatrix}.$$

Here D_N is a block operator, not the scalar reference twist $D_{N,b}$. Suppose, uniformly on U and in N , that

$$\|D_N + tE_N\| \leq q \quad (0 \leq t \leq 1), \quad \|D_N\|_{C^k(\mathfrak{S}_1)} \leq M_k, \quad \|E_N\|_{C^k(\mathfrak{S}_2)} \leq \varepsilon_N \leq 1.$$

Suppose also that there are functions ℓ_-, ℓ_+ with bounded C^k norms such that

$$\|\log \det(I + T_{N,\pm}) - \ell_{\pm}\|_{C^k(U)} \leq \delta_N.$$

All logarithms are defined by the power series at zero. Then

$$\|\log \det(I + T_N) - \ell_- - \ell_+\|_{C^k(U)} \leq C_{k,q,M_k}(\delta_N + \varepsilon_N^2). \quad (2.16)$$

The constant is independent of the dimensions of both end spaces. If T_N is first replaced by $\tilde{T}_N$ with $C^k(\mathfrak{S}_1)$ error η_N , with the same operator margin and bounded $C^k(\mathfrak{S}_1)$ norms along the intervening segment, an additional $C\eta_N$ suffices on the right.

Proof. For $\|T\| < 1$ put

$$\mathcal{L}(T) = \sum_{h \geq 1} \frac{(-1)^{h+1}}{h} \operatorname{tr}(T^h).$$

This is the indicated logarithm, whether or not the matrix is symmetric. Along $T(t) = D + tE$, let $R(t) = (I + D + tE)^{-1}$. Differentiating the convergent series, and then using cyclicity of trace, gives

$$\frac{d}{dt} \mathcal{L}(D + tE) = \operatorname{tr}(R(t)E), \quad \frac{d^2}{dt^2} \mathcal{L}(D + tE) = -\operatorname{tr}(R(t)ER(t)E).$$

The inverse $R(0)$ is block diagonal and E is off diagonal. Consequently $\operatorname{tr}(R(0)E) = 0$. Taylor's formula is therefore the exact identity

$$\mathcal{L}(D + E) - \mathcal{L}(D) = - \int_0^1 (1-t) \operatorname{tr}(R(t)ER(t)E) dt. \quad (2.17)$$

Since $\|R(t)\| \leq (1-q)^{-1}$ and $|\operatorname{tr}(AB)| \leq \|A\|_2 \|B\|_2$, the undifferentiated absolute value is at most $\|E\|_2^2/[2(1-q)^2]$. For parameter derivatives use $\partial R = -R\partial(D + tE)R$ repeatedly. Each differentiated integrand in (2.17) still contains the two displayed E factors, possibly differentiated. Place these in $\mathfrak{S}_2$ and all other factors in operator norm. The bounds on D , E and R yield $C_{k,q,M_k}\varepsilon_N^2$. The number of product-rule terms depends on k , not on dimension. The block-diagonal determinant is the product of its two determinants; adding their stipulated errors proves (2.16).

For the last assertion write $T(t) = T + t(\tilde{T} - T)$ and integrate $\operatorname{tr}((I + T(t))^{-1}(\tilde{T} - T))$. In every parameter derivative one factor is a derivative of $\tilde{T} - T$ in trace norm; all remaining factors have uniformly bounded operator norm. This gives $C\eta_N$. These arguments also justify each stated fixed derivative: all the finite-dimensional identities have bounds uniform in dimension. $\square$

A convenient sufficient condition for the diagonal hypothesis is $C^k(\mathfrak{S}_1)$ convergence of the padded diagonal blocks to trace-class half-line operators of norm below q . The same path argument gives convergence of their logarithms. Thus no determinant estimate proportional to the number of interior collisions is needed. The quadratic off-diagonal term expresses the fact that a closed trace must cross from one end to the other and back. A first-order trace-norm estimate remains sufficient when this refinement is not needed.

Corollary 2.8 (Relative comparison before conditioning). *Let Q be a fixed compact gate and let the unconditioned endpoint density be exactly*

$$w_N = c d_N \beta_N r_N, \quad c > 0, \quad d_N > 0.$$

Assume uniform positive lower and upper bounds for β_N, r_N and their limits β, r , uniform C^k bounds, and

$$\|\log \beta_N - \log \beta\|_{C^k(Q)} + \|r_N - r\|_{C^k(Q)} \leq e_N.$$

If $Z_N = \int_Q \beta_N r_N$ and $Z = \int_Q \beta r$, then

$$\left\| \frac{w_N}{\int_Q w_N} - \frac{\beta r}{Z} \right\|_{C^k(Q)} + \left| \frac{\int_Q w_N}{c d_N} - Z \right| \leq C e_N. \quad (2.18)$$

The constant is independent of d_N , which may tend to zero at any rate. In particular, a scalar cofactor identity $\log \beta_N = b_N - \log \det(I + T_N)$, with a separated edge sum b_N and the hypotheses of Proposition 2.7, gives such a comparison with the corresponding end errors.

Proof. Exponentiation is locally Lipschitz on bounded C^k sets. Products therefore compare $\beta_N r_N$ with βr with error $C e_N$. Integration compares Z_N with Z , and the fixed positive gate gives a positive lower bound for both integrals. Division by these integrals proves the first estimate. The second uses the exact equality $\int_Q w_N = c d_N Z_N$. Neither step divides an approximation error by d_N . $\square$

For the billiard operator the end spaces are the first and last $\lfloor N/3 \rfloor$ sites after the middle perturbation has been removed. The Green and gluing bounds in the proof of Theorem 2.3 give the trace-norm deletion error, the two diagonal limits and the Hilbert–Schmidt off-diagonal bound, all exponentially small up to fixed polynomial factors in N . Shrinking the collar gives the operator margin along the entire segment. Consequently the proposition applies before the cofactor is converted to a probability law. In Corollary 2.8 one takes $d_N = D_{N,b}$, $c = (2\pi\mathcal{A})^{-1}$, and $r_N = d - E_{N,b} + p_b u - p_b v$. The geometry is responsible for these hypotheses and for the exact physical clock; the operator comparison itself does not use periodicity, a billiard reflection rule, or a lower bound on the total success mass.

3. CONTACT RECONSTRUCTION ALONG A MARKED PERIODIC ORBIT

The future and past actions of a periodic orbit need not coincide. Two positive offsets nevertheless separate them in the physical window law, without a supplied flux model or time-reversal symmetry. In the measured tangent coordinates, the derivative of the future action is a sum of successive boundary visits. We identify its signed cyclic homogeneous blocks and prove the corresponding finite-jet factorization for actual smooth boundaries. The next section recovers the positive quadratic parameters globally; Section 5 then separates the actual smooth functions, including flat differences. The inverse on analytic function spaces is a separate extension in the full technical manuscript, Theorem F.3.6.

3.1. Fixed geometric marks and the observation coordinates. Fix a clear nongrazing periodic orbit with $r \geq 2$ collision phases. The collision points are distinct modulo the prescribed lattice, and phase i has a fixed point q_i , unit outward obstacle normal n_i and signed unit tangent t_i . These data, including the lifted closing translation, will be called its *marked polygon*. Indices are read modulo r , with that translation understood. Set

$$e_i^0 = \frac{q_{i+1} - q_i}{L_i}, \quad L_i = |q_{i+1} - q_i|, \quad \nu_i = e_i^0 \cdot n_i > 0, \quad p_i = e_i^0 \cdot t_i / \nu_i.$$

The reflection condition is part of the marked polygon. In particular $e_{i-1}^0 \cdot n_i = -\nu_i$ and $e_{i-1}^0 \cdot t_i = e_i^0 \cdot t_i$. No curvature or higher boundary derivative is included in these marks.

Write the unknown arcs as normalized normal graphs

$$R_i(x) = q_i + \nu_i^{-1} \{t_i x - n_i F_i(x)\}, \quad F_i(0) = F_i'(0) = 0, \quad c_i := F_i''(0) > 0. \quad (3.1)$$

The actual curvature is $\kappa_i = \nu_i c_i$. The recorded coordinate is the scaled tangent projection $x = \nu_i t_i \cdot (R_i - q_i)$, not the unknown normal displacement. Thus recording x does not supply $F_i(x)$. The marks specify the coordinate, and its use in the inverse does not require converting measured arclength by means of an unknown boundary. Put $k_i = L_i^{-1}$ and

$$\ell_i(x, y) = |R_{i+1}(y) - R_i(x)|, \quad h_i(x, y) = \ell_i(x, y) - L_i + p_i x - p_{i+1} y.$$

These gauges have zero value and differential at the origin. The marked polygon remains a stationary orbit for every anchored graph in (3.1). On a sufficiently small common real collar, small perturbations preserve positive curvature, clearance and nongrazing incidence.

Lemma 3.1 (Signed Jacobi coordinates). *There are signs $\epsilon_i \in \{1, -1\}$, determined by the marked polygon and tangent choices, such that*

$$D^2 h_i(0, 0) = \begin{pmatrix} k_i + c_i & -\epsilon_i k_i \\ -\epsilon_i k_i & k_i + c_{i+1} \end{pmatrix}. \quad (3.2)$$

On the infinite itinerary a diagonal change of signs makes all mixed entries negative. The resulting Jacobi operator has positive masses $2c_i$. The future half-line with endpoint $x_0 = u$ has linearization

$$x'_{b,j}(0) = \sigma_b \sigma_{b+1} \cdots \sigma_{b+j-1}, \quad \sigma_i = \epsilon_i a_i, \quad 0 < a_i < 1. \quad (3.3)$$

The factors a_i are periodic. They depend smoothly on the positive curvatures, and are uniformly bounded away from one on a compact marked class. In particular $\Lambda := \prod_{i=0}^{r-1} \sigma_i$ satisfies $|\Lambda| < 1$.

The relative and physical laws of Section 2 hold in the graph coordinates, for a returning oriented period P (which may be taken to be $2r$). In these coordinates write their limits as $S_b^\pm, B_b^\pm$ and

$$A_b(u) = S_b^-(u) - p_b u, \quad C_b(v) = S_b^+(v) + p_b v. \quad (3.4)$$

Then the physical residual in the limiting law is $d - A_b(u) - C_b(v)$. The action and amplitude estimates remain estimates at each separately fixed differentiation order.

Proof. At the origin $R'_i = t_i/\nu_i$ and $R''_i = -n_i c_i/\nu_i$. Differentiating a chord twice gives the same-end entry $k_i + c_i$. Its mixed entry is

$$-\frac{t_i \cdot t_{i+1} - (e_i^0 \cdot t_i)(e_i^0 \cdot t_{i+1})}{L_i \nu_i \nu_{i+1}}.$$

The two transverse projections have lengths ν_i, ν_{i+1} ; their relative sign defines ϵ_i . This proves (3.2). If $s_0 = 1$ and $s_{j+1} = \epsilon_{b+j} s_j$, the substitution $x_j = s_j y_j$ gives the positive-mass operator

$$(Hy)_j = (k_{b+j-1} + k_{b+j} + 2c_{b+j})y_j - k_{b+j-1}y_{j-1} - k_{b+j}y_{j+1}.$$

Its Green kernel and fixed differentiated bounds follow from the walk expansion in the preceding section; the diagonal change of signs preserves their absolute bounds.

For endpoint $y_0 = 1$, finite Dirichlet solutions are strictly positive in their interiors by the discrete maximum principle. Their half-line limit is positive and decays exponentially. Define $z_{j+1} = k_{b+j}(y_{j+1} - y_j)$. The recurrence gives $z_{j+1} - z_j = 2c_{b+j}y_j$. Since $z_j \rightarrow 0$ at infinity, $z_{j+1} = -\sum_{h \geq j+1} 2c_{b+h}y_h < 0$. Consequently $0 < y_{j+1} < y_j$. Uniqueness of a decaying tail shows that its ratio at a site depends only on that phase; call it a_i . This proves (3.3). Smooth dependence follows by differentiating the uniformly invertible weighted equations. Compactness and finitely many phases give $\max_i a_i < 1$ uniformly.

The signed charts return after $2r$ steps, even when they do not return after r . This does not create $2r$ independent unknown contact germs. To compare with signed scaled arclength at the

initial phase, let $\phi_b(x)$ be its expression in the graph coordinate, with the same initial orientation. Directly from (3.1),

$$\phi_b(x) = \int_0^x \sqrt{1 + F'_b(s)^2} ds = x + O(x^3).$$

Use the positive branch on the real collar. The endpoint twist is multiplied by $\phi'_b(u)\phi'_b(v)$, whose value at zero is one. The exact reference $D_{N,b}$ is therefore unchanged. The graph future action is the arclength action composed with ϕ_b , plus $p_b(x - \phi_b(x))$; the past action has the opposite gauge correction. The amplitude acquires the ordinary endpoint Jacobian. In particular the graph actions are still anchored and have positive Hessians at zero. Composition on smaller collars preserves the fixed-order relative bounds. The physical actions in (3.4) are simply the physical arclength actions composed with ϕ_b ; no physical time has been changed. This proves the last assertion, without making an unknown coordinate conversion an input to the reconstruction. $\square$

3.2. Separation of unequal endpoint actions. Choose two fixed offsets $0 < d_1 < d_2$ in the physical positive window, and a common square I^2 on which both residuals have a positive floor. At phase b the two limiting densities are

$$f_{b,d}(u, v) = Z_{b,d}^{-1} B_b^-(u) B_b^+(v) \{d - A_b(u) - C_b(v)\}, \quad d = d_1, d_2. \quad (3.5)$$

Here $A_b(0) = C_b(0) = 0$ and $B_b^\pm(0) = 1$. The two normalizing constants are unrelated unknown positive numbers.

Proposition 3.2 (Two-offset action separation). *The two conditional laws in (3.5) determine the complete future and past actions and normalized amplitudes. More precisely, suppressing the phase index, set*

$$Q(u, v) = \frac{f_{d_1}(u, v) f_{d_2}(0, 0)}{f_{d_2}(u, v) f_{d_1}(0, 0)}, \quad H(u, v) = \frac{d_1 d_2 \{1 - Q(u, v)\}}{d_2 - d_1 Q(u, v)}. \quad (3.6)$$

Then

$$A(u) = H(u, 0), \quad C(v) = H(0, v), \quad H(u, v) = A(u) + C(v). \quad (3.7)$$

The gauged actions follow from (3.4), and

$$B^-(u) = \frac{f_d(u, 0)}{f_d(0, 0)} \frac{d}{d - A(u)}, \quad B^+(v) = \frac{f_d(0, v)}{f_d(0, 0)} \frac{d}{d - C(v)}. \quad (3.8)$$

For each fixed $M \geq 0$ these formulas are locally Lipschitz in $C^M(I^2)$, on compact positive-density classes with fixed separated offsets and residual floors. They do not require equality of the two actions, a model for the amplitudes, or an acceptance count.

Proof. Put $h = A(u) + C(v)$. Cancellation of the amplitudes and the two normalizers gives

$$Q = \frac{1 - h/d_1}{1 - h/d_2}, \quad d_2 - d_1 Q = \frac{d_2(d_2 - d_1)}{d_2 - h} > 0.$$

Solving the first identity for h proves (3.6)–(3.7). Since h is bounded on the common square, the second identity gives a positive denominator floor, uniformly on the stated compact class. Taking a single density ratio on each coordinate axis proves (3.8). Continuous densities are fixed everywhere on the positive square by their probability laws, even though finite samples would require estimation.

For stability, slice evaluation is bounded in C^M . Products and reciprocals are locally Lipschitz in C^M when the denominators have positive floors. Apply this first to Q , then to H , and finally to the two axis formulas. Small perturbations need not themselves satisfy the separation identity. The formulas remain defined on an open neighborhood with the same, possibly halved,

denominator floors. The constants are not uniform as $d_2 - d_1$ or a residual floor tends to zero. No assertion of a minimal number of offsets is made. $\square$

3.3. Actual boundary variations and cyclic blocks. Let $\mathcal{S}(F) = (S_0^-, \dots, S_{r-1}^-)$ be the future action map in the fixed graph coordinates. A reference half-line starting at phase b is denoted $x_{b,j}(u)$, and its j th unit chord direction is $e_{b,j}(u)$. Define

$$\alpha_b(u) = \frac{e_{b,0}(u) \cdot n_b}{\nu_b}, \quad v_{b,j}(u) = \frac{(e_{b,j}(u) - e_{b,j-1}(u)) \cdot n_{b+j}}{\nu_{b+j}} \quad (j \geq 1). \quad (3.9)$$

At zero these weights are respectively 1 and 2.

Lemma 3.3 (Stationary boundary visits). *For an anchored smooth variation $\eta = (\eta_i)$, on a sufficiently small common collar the derivative of the actual action is*

$$(D\mathcal{S}(F)\eta)_b(u) = \alpha_b(u)\eta_b(u) + \sum_{j \geq 1} v_{b,j}(u)\eta_{b+j}(x_{b,j}(u)). \quad (3.10)$$

The series and its stipulated fixed derivatives converge. The action jet through degree M depends only on the graph jets through degree M , for every $M \geq 2$. This is a statement about smooth representatives, not only formal Taylor series.

Proof. The weighted Green construction of the preceding section applies to the signed graph equations. It gives the actual stationary half-line and bounds $|x_{b,j}(u)| \leq C|u|\rho^j$, with the corresponding bounds for each fixed differentiated order. Changing a graph by η_i changes its position by $-n_i\eta_i/\nu_i$. Direct differentiation of one chord therefore contributes

$$\frac{e_{b,j} \cdot n_{b+j}}{\nu_{b+j}} \eta_{b+j}(x_{b,j}) - \frac{e_{b,j} \cdot n_{b+j+1}}{\nu_{b+j+1}} \eta_{b+j+1}(x_{b,j+1}).$$

The marked linear gauges have zero variation. Differentiate the sum of the first J gauged chords along the stationary family. All internal orbit variations cancel. The initial one is zero because its coordinate is fixed. Retain the terminal term

$$h_{b+J-1,2}(x_{b,J-1}, x_{b,J}) D_F x_{b,J}[\eta].$$

The first factor is $O(|u|\rho^{J-1})$, since $Dh(0,0) = 0$; the second decays exponentially by the differentiated weighted equations. Thus this term, and each of its separately fixed derivatives, tends to zero. Direct graph variations are summable, because anchored variations are $O(x^2)$. Gathering the initial visit once and every internal visit twice gives precisely (3.10).

For completeness, suppose two smooth profiles have the same jets through M . Their difference obeys $\eta_i(x) = O(x^{M+1})$ on a smaller collar. Interpolate the profiles there. Their positive quadratic jets agree, and uniform small-collar estimates hold along the interpolation; no global interpolation of obstacles is needed. The envelope formula gives, uniformly along this path,

$$|D\mathcal{S}(F)[\eta]_b(u)| \leq C|u|^{M+1} \left(1 + \sum_{j \geq 1} \rho^{j(M+1)} \right).$$

Integration along the path shows that the two actual actions differ by $O(|u|^{M+1})$. Their smoothness, already obtained from the weighted equations, implies equality of the Taylor jets through M . This jet argument alone does not identify equality of all smooth jets with equality of smooth germs. The latter identification from the full action functions is established by the additional real weighted estimate in Section 5. $\square$

For a degree $n \geq 2$ define the signed cyclic shift on $\mathbb{R}^r$ by

$$(T_n z)_b = \sigma_b^n z_{b+1}, \quad T_n^r = \Lambda^n I, \quad \mathcal{C}_n = (I + T_n)(I - T_n)^{-1}. \quad (3.11)$$

The notation uses the physical r contact germs, not the doubled list of oriented charts.

Theorem 3.4 (All-order periodic contact blocks). *Let $q_n = (F_b^{(n)}(0))_b$ and $s_n = ((S_b^-)^{(n)}(0))_b$. The map $q_2 \mapsto s_2$ is locally invertible throughout the positive-curvature class; its derivative is $\mathcal{C}_2$. For every $n \geq 3$,*

$$s_n = \mathcal{C}_n q_n + \mathcal{R}_n(q_2, \dots, q_{n-1}), \quad q_n = (I - T_n)(I + T_n)^{-1}(s_n - \mathcal{R}_n). \quad (3.12)$$

The coefficients also depend on the fixed marked polygon. The formulas hold for actual smooth graphs. They give a local inverse through every finite order, without supplied curvatures.

In the maximum row-sum norm, with $a_ := \max_b a_b < 1$,*

$$\max(\|\mathcal{C}_n\|_\infty, \|\mathcal{C}_n^{-1}\|_\infty) \leq \frac{1 + a_*^n}{1 - a_*^n} \quad (n \geq 2), \quad \det \mathcal{C}_n = \frac{1 - (-1)^r \Lambda^n}{1 - \Lambda^n} > 0. \quad (3.13)$$

In particular the determinant is one for an even contact cycle. For an odd cycle it need not be one, but is never zero. These bounds concern the homogeneous blocks, not arbitrary lower-degree smooth couplings.

Proof. Vary only the new degree by $\eta_i(x) = z_i x^n / n!$, using a smooth cutoff equal to one near zero when necessary. In (3.10) the coefficient of $u^n / n!$ uses only $\alpha_b(0) = 1$, $v_{b,j}(0) = 2$, and (3.3). It is therefore

$$z_b + 2 \sum_{j \geq 1} (\sigma_b \cdots \sigma_{b+j-1})^n z_{b+j} = \left(I + 2 \sum_{j \geq 1} T_n^j \right) z = (I + T_n)(I - T_n)^{-1} z.$$

This proves the derivative formula in degree two. For $n \geq 3$, the linearized orbit and the constant visit weights depend only on the quadratic jets, not on the fresh degree. Together with the smooth-jet factorization of Lemma 3.3, integration in that fresh degree gives the affine identity (3.12).

Since $\|T_n\|_\infty \leq a_*^n < 1$, the two geometric series for $(I - T_n)^{-1}$ and $(I + T_n)^{-1}$ converge. They give the inverse and both estimates in (3.13). For a cyclic shift, $\det(I - zT_n) = 1 - z^r \Lambda^n$. Apply this at $z = 1, -1$ to obtain the displayed determinant; $|\Lambda| < 1$ makes its numerator and denominator positive. The finite-dimensional inverse function theorem now inverts the degree-two map locally. Successive applications of (3.12) invert all later degrees. At any fixed order this is a smooth inverse on an open finite-jet neighborhood; its constants can be chosen uniformly on compact subsets of that neighborhood. No global uniqueness of the degree-two map is needed or asserted. $\square$

For a normal two-contact channel, the positive signed factors are $\sigma_b = \tau_b e^{-\gamma}$ and $\Lambda = e^{-2\gamma}$. Substitution in (3.11) gives

$$\mathcal{C}_n = \begin{pmatrix} \coth(n\gamma) & \tau_0^n \operatorname{csch}(n\gamma) \\ \tau_1^n \operatorname{csch}(n\gamma) & \coth(n\gamma) \end{pmatrix}.$$

Thus the earlier determinant-one inverse is the two-site specialization. For an odd cycle, replacing the signed σ_i by their absolute values would generally change the odd-degree reconstruction. Doubling the oriented forward period does not justify making that replacement.

4. GLOBAL IDENTIFICATION OF THE POSITIVE QUADRATIC PARAMETERS

The preceding finite-jet calculation does not by itself exclude a second positive quadratic branch. We resolve this question directly, without an analytic inverse theorem. A Schur-complement identity converts the curvature inverse into a strict contraction with constants determined by the observed action Hessians. This proves global identification of the positive quadratic parameters, characterizes their image and gives an a posteriori residual estimate. Together with the signed higher-order blocks, it supplies the finite-jet input for the actual smooth inverse in Section 5; quadratic identification alone is not asserted to identify a complete smooth germ.

Throughout this section the polygon, graph coordinates and signs are those of Section 3. In particular $k_i = L_i^{-1} > 0$, $c_i = F_i''(0) > 0$ and $s_i = (S_i^-)''(0)$; indices are modulo the physical contact period r . We regard k as fixed and write $\mathcal{R}_k(c) = s$. This is a map of quadratic data, independent of the higher contact jets. Its definition uses the actual half-line construction, not an arbitrary solution of a Riccati equation.

4.1. The quadratic inverse.

Lemma 4.1 (Boundary Schur complement). *For every positive curvature vector c , the actual half-line action Hessians satisfy $0 < c_i < s_i$ and*

$$s_i = k_i + c_i - \frac{k_i^2}{k_i + c_{i+1} + s_{i+1}}, \quad a_i = \frac{k_i}{k_i + c_{i+1} + s_{i+1}} \in (0, 1). \quad (4.1)$$

Here a_i is the unsigned one-step contraction in Lemma 3.1. Conversely, positive vectors c, s satisfying the first identity have $s = \mathcal{R}_k(c)$.

Proof. The half-line starting at i is obtained by adjoining its first edge to the half-line starting at $i + 1$. At quadratic order its stationary value is the minimum over y of

$$\frac{1}{2}(k_i + c_i)u^2 - \epsilon_i k_i u y + \frac{1}{2}(k_i + c_{i+1} + s_{i+1})y^2.$$

The strictly positive half-line Hessian established in the preceding section gives $s_{i+1} > 0$. Eliminating y proves the first identity; the stationary ratio is $y/u = \epsilon_i a_i$. The denominator exceeds k_i , proving both $a_i < 1$ and $s_i > c_i$.

For the converse, define a_i by the second identity. Substituting the first identity at $i + 1$ gives

$$\frac{k_i}{a_i} = k_i + k_{i+1} + 2c_{i+1} - k_{i+1}a_{i+1}.$$

Thus the products $y_j = a_i \cdots a_{i+j-1}u$ solve the sign-conjugated Jacobi equation with boundary value u . They decay exponentially because there are finitely many phases and every $a_i < 1$. Coercivity makes this decaying solution unique: the difference of two such solutions has zero boundary value, and summation of its positive Jacobi quadratic form gives zero. Restoring the signs gives the actual linearized half-line. Its initial action Hessian is $k_i + c_i - k_i a_i = s_i$. Consequently these are the actual action Hessians. $\square$

For $s \in (0, \infty)^r$ and $z \in [0, \infty)^r$, put

$$(\Phi_s z)_i = \left[s_i - k_i + \frac{k_i^2}{k_i + s_{i+1} + z_{i+1}} \right]_+, \quad [t]_+ = \max(t, 0), \quad (4.2)$$

$$q(s) = \max_i \left(\frac{k_i}{k_i + s_{i+1}} \right)^2 < 1.$$

The positive-part operation specifies an inverse algorithm even for positive data that are not realizable by strictly positive curvatures. It does not add zero-curvature tables to the geometric class.

Theorem 4.2 (Global curvature coordinates). *The map $\mathcal{R}_k : (0, \infty)^r \rightarrow (0, \infty)^r$ is one-to-one and a real analytic diffeomorphism onto its open image. That image and the inverse admit the following description.*

For every $s > 0$, the map Φ_s has a unique fixed point $z(s)$ in the closed positive orthant. Its iterates from any $z^{(0)} \geq 0$ converge to $z(s)$, with

$$\|z^{(m)} - z(s)\|_\infty \leq \frac{q(s)^m}{1 - q(s)} \|z^{(1)} - z^{(0)}\|_\infty \quad (m \geq 0). \quad (4.3)$$

The first-increment estimate (4.3) is an a priori bound. A residual at the current iterate gives the a posteriori bound

$$\|z^{(m)} - z(s)\|_\infty \leq E_m(s), \quad E_m(s) := \frac{\|\Phi_s z^{(m)} - z^{(m)}\|_\infty}{1 - q(s)}. \quad (4.4)$$

For exact iteration its numerator is $\|z^{(m+1)} - z^{(m)}\|_\infty$. Precisely,

$$s \in \mathcal{R}_k((0, \infty)^r) \iff z_i(s) > 0 \text{ for every } i; \quad \mathcal{R}_k^{-1}(s) = z(s). \quad (4.5)$$

For any $s, \tilde{s} > 0$, define the observable constant

$$q_* = \max_i \left(\frac{k_i}{k_i + \min(s_{i+1}, \tilde{s}_{i+1})} \right)^2 < 1.$$

Then, including for data outside the geometric image,

$$\|z(s) - z(\tilde{s})\|_\infty \leq \frac{1 + q_*}{1 - q_*} \|s - \tilde{s}\|_\infty. \quad (4.6)$$

Proof. For $z \geq 0$ the expression inside the positive part is strictly smaller than s_i . Hence Φ_s maps the closed orthant into $\prod_i [0, s_i]$. Its untruncated expression has derivative in z_{i+1} equal to $-k_i^2/(k_i + s_{i+1} + z_{i+1})^2$, of absolute value at most $q(s)$. The positive part is one-Lipschitz. The map is therefore a strict contraction in the maximum norm on the complete closed orthant. The contraction theorem and summation of the successive differences prove existence, uniqueness and (4.3). For any $v \geq 0$, the triangle inequality and contraction give

$$\|v - z(s)\|_\infty \leq \|v - \Phi_s v\|_\infty + q(s) \|v - z(s)\|_\infty.$$

This proves (4.4). It also provides a bound for an inexact evaluation: if w satisfies $\|w - \Phi_s v\|_\infty \leq \eta$, replace the residual numerator by $\|w - v\|_\infty + \eta$. An error bound for evaluating the map, when used, is part of this certificate and is not assumed to be zero.

If $s = \mathcal{R}_k(c)$ for $c > 0$, Lemma 4.1 says that $c = \Phi_s c$. Therefore $c = z(s)$, which proves global injectivity. If $z(s) > 0$, no positive part is active at the fixed point, so (4.1) holds with $c = z(s)$. The converse in that lemma proves the other implication in (4.5). This proves the image characterization without identifying formal positive data with geometric data.

For the two-point estimate, changing s_i contributes at most $\|s - \tilde{s}\|_\infty$ to $\Phi_s z$; changing s_{i+1} in its denominator contributes at most $q_* \|s - \tilde{s}\|_\infty$. All intermediate denominators are bounded below by the one used in q_* . Changing z contributes at most $q_* \|z - \tilde{z}\|_\infty$. Apply these bounds at the two fixed points and move the last term to the left. This proves (4.6).

On the image the fixed point is strictly positive. Locally the positive part can be omitted, and the remaining equation is real analytic in (s, c) . Its derivative in c is $I + T_2$, where $(T_2 v)_i = a_i^2 v_{i+1}$, and has a convergent Neumann inverse because $\|T_2\|_\infty < 1$. Thus the inverse is locally real analytic. The forward half-line Hessian is real analytic in c : for example the positive half-line Jacobi operator with periodic coefficients has a locally uniformly convergent resolvent expansion,

and the boundary Schur complement is its analytic matrix element. Differentiating (4.1) also gives

$$(I - T_2) ds = (I + T_2) dc. \quad (4.7)$$

Both factors are invertible. Hence the forward map is a local real analytic diffeomorphism at every $c > 0$. Its image is open, and the already proved injectivity makes these local inverses one global inverse on that image. $\square$

The projection in (4.2) is useful only in the reconstruction algorithm; it plays no role in the analytic inverse near a realized datum. Formula (4.3) also gives a finite sufficient test for membership in the image: if every component of an iterate exceeds either certified error bound, then $z(s) > 0$. For the residual bound this test is $\min_i z_i^{(m)} > E_m(s)$. It certifies membership in the quadratic image, not extension to a prescribed collection of globally disjoint analytic obstacles. The constants are allowed to deteriorate as an action Hessian tends to zero. No uniform assertion at a grazing or zero-curvature limit is used.

There is a second direct uniqueness check. If two positive curvature vectors have the same s , subtraction of (4.1) gives

$$c_i - \tilde{c}_i = -a_i \tilde{a}_i (c_{i+1} - \tilde{c}_{i+1}).$$

Iteration around the cycle forces the difference to vanish, since the product of the $a_i \tilde{a}_i$ is strictly smaller than one. This argument treats odd and even physical periods alike. It uses only quadratic data; the signs in odd higher contact orders still have to be retained in the geometric reconstruction.

For a normal two-contact channel, $k_0 = k_1 = 1/g$ and c_i are the physical curvatures. Set $H = (1 + g^2 s_0 s_1)^{1/2}$. The recovered curvatures are

$$c_0 = \frac{H \sqrt{s_0/s_1} - 1}{g}, \quad c_1 = \frac{H \sqrt{s_1/s_0} - 1}{g}.$$

These are precisely the alternating curvature formulas of Theorem F.23.1. Indeed put $u_i = g s_i$ and $C_i = 1 + g c_i$; then $C_0 C_1 = 1 + u_0 u_1$ and $C_0/C_1 = u_0/u_1$, which give $(C_0 - u_0)(C_1 + u_1) = 1$ and its reversed version. These are the two identities in (4.1). Whenever the displayed curvatures are positive, uniqueness identifies them with the fixed point. This is a specialization of the geometric inverse, not a reduction of the two observation models.

Not every positive vector is an action datum for positive curvatures. For example $r = 2$, $k_0 = k_1 = 1$ and $s = (1/10, 10)$ give $z(s) = (0, 109/11)$. The first component of the untruncated map is negative for every $z \geq 0$, and the second fixed-point equation gives the stated value. Thus this s is excluded by (4.5), even though the stable reconstruction algorithm is defined there.

5. SMOOTH CONTACT RIGIDITY FROM THE COMPLETE BOUNDARY LAW

The boundary law determines functions on a real collar, not merely their Taylor coefficients. This distinction removes the analytic hypothesis from local contact identification. After the finite jets have been identified, the actual stationary envelope separates the remaining functions: its initial evaluation is invertible, whereas every later evaluation takes place strictly nearer the contact. A weighted real norm makes the latter part a contraction. The argument uses neither convergence of Taylor series nor a holomorphic extension.

Throughout this section the polygon, the physical phase labels and the graph sensors are those of Section 3. We write $\mathcal{S}(F) = (S_b^-)_{b=0}^{r-1}$ for the actual future actions, with the same anchored gauges. Contact identification below means equality on a possibly smaller real neighborhood of each marked contact. Continuation to unvisited portions of a smooth obstacle is not part of that conclusion.

5.1. A common real collar and successive visits.

Lemma 5.1 (Uniform real visits). *Let F^t , $0 \leq t \leq 1$, be a smooth compact family of anchored C^∞ graph tuples with positive quadratic jets, in the fixed marked coordinates. There are $R > 0$ and $a < 1$ such that the actual future half-lines are defined for $|u| \leq R$ and all t , and*

$$x_{b,j}^t(u) = X_{b+j-1}^t \circ \cdots \circ X_b^t(u), \quad |x_{b,j}^t(u)| \leq a^j |u|, \quad j \geq 1, \quad (5.1)$$

where $X_i^t(u) = x_{i,1}^t(u)$. On this collar the real envelope weights of (3.9) satisfy

$$\frac{1}{2} \leq \alpha_b^t(u) \leq \frac{3}{2}, \quad 1 \leq v_{b,j}^t(u) \leq 3 \quad (j \geq 1). \quad (5.2)$$

The constants can be chosen uniformly on bounded finite-smoothness classes with fixed positive curvature bounds and fixed nongrazing marks. For each stipulated differentiation order only finitely many bounds for the graph derivatives are needed.

Proof. The positive Jacobi construction and its weighted implicit equations in Sections 2 and 3 give the unique decaying stationary half-line. Their inverses are uniform on compact positive quadratic classes. Applying the construction with the parameter t gives a common collar and continuous first derivatives of X_i^t there. By Lemma 3.1,

$$X_i^t(0) = 0, \quad (X_i^t)'(0) = \sigma_i(t), \quad a_0 := \max_{i,t} |\sigma_i(t)| < 1.$$

Fix $a_0 < a < 1$ and shrink the collar until $\sup_{i,t,|u| \leq R} |(X_i^t)'(u)| \leq a$. A shifted tail solves the same decaying stationary problem at the next physical phase. Uniqueness therefore identifies it with that phase's half-line. Iteration proves the first identity in (5.1); the mean value theorem then proves the second. In particular the multiplicative constant is one. The weaker estimate $C\rho^j|u|$ alone would not suffice for the large-weight argument below when $C\rho > 1$.

Each unit chord is a smooth function of two contact coordinates on a common neighborhood of the clear reference chord. There are only finitely many physical phases. At the origin the two weights are 1 and 2, respectively. Uniform continuity, followed by a further shrink of R , gives (5.2) for every incident pair of chords, hence for every j . Signed tangent choices have not been changed; these are normal projection weights of actual chords. For a bounded finite-smoothness class, the same proof uses compactness of its positive quadratic range and uniform bounds in the weighted implicit equations. Bounds through any fixed order follow by successive differentiation with a strict exponential margin. No uniform assertion as curvature or the nongrazing margin vanishes is used. $\square$

For an integer $m \geq 2$ define the weighted real norm

$$\|h\|_{m,R} = \max_b \sup_{0 < |u| \leq R} \frac{|h_b(u)|}{|u|^m}. \quad (5.3)$$

It is finite for a C^m tuple whose derivatives of orders less than m vanish at zero. It is a norm on such functions, not a norm of their formal series.

Proposition 5.2 (Separation of the nonlinear remainder). *Suppose that two smooth graph tuples F, G have the same contact jets through order $m - 1$. Apply Lemma 5.1 to their segment $F^t = F + t(G - F)$ on a common collar. If*

$$\theta_m := \frac{6a^m}{1 - a^m} < 1, \quad (5.4)$$

then

$$\|G - F\|_{m,R} \leq \frac{2}{1 - \theta_m} \|\mathcal{S}(G) - \mathcal{S}(F)\|_{m,R}. \quad (5.5)$$

The same assertion holds with finite C^m differences whenever the real half-lines and envelope used here have the indicated regularity.

Proof. Put $h = G - F$ and $g = \mathcal{S}(G) - \mathcal{S}(F)$. Integrate the actual envelope of Lemma 3.3 along the segment:

$$g_b(u) = \bar{\alpha}_b(u)h_b(u) + (Kh)_b(u), \quad \bar{\alpha}_b(u) = \int_0^1 \alpha_b^t(u) dt, \quad (5.6)$$

where the linear operator K , with this segment fixed, is

$$(Kh)_b(u) = \int_0^1 \sum_{j \geq 1} v_{b,j}^t(u) h_{b+j}(x_{b,j}^t(u)) dt.$$

The operator is applied to the actual difference, not to a freely chosen formal variation. The initial coefficient is independent of h_b in this displayed identity once the segment is fixed. For any function in the weighted space,

$$|h_{b+j}(x_{b,j}^t(u))| \leq \|h\|_{m,R} a^{jm} |u|^m.$$

This summable majorant justifies integration and summation and gives

$$\|Kh\|_{m,R} \leq \frac{3a^m}{1-a^m} \|h\|_{m,R}, \quad \|\bar{\alpha}^{-1}\|_{\infty} \leq 2.$$

Multiplication by $\bar{\alpha}^{-1}$ in (5.6) and the triangle inequality prove (5.5). Alternatively, the inverse of $I + \bar{\alpha}^{-1}K$ is its norm-convergent Neumann series. The coefficients and composition maps of that operator depend on both candidates; this proof does not present that pair-dependent inverse as an algorithm using only the observations.

The finite-jet factorization already proved for actual smooth graphs shows that g has the same vanishing order as required for this norm. The terminal contribution in the finite envelope has already been sent to zero with uniform parameter bounds before the identity is integrated. It is not dropped by declaring a formal half-line limit. $\square$

5.2. Identification of smooth germs.

Theorem 5.3 (Smooth marked contact rigidity). *Fix a clear nongrazing marked polygon with distinct physical contacts. Two positive-curvature C^∞ contact tuples with the same phase-resolved two-offset laws (3.5) have identical contact germs in the measured graph coordinates. No analyticity, curvature input, separately calibrated local flux amplitude or closeness between the candidates is required. In fact equality of their actual future action germs at all physical phases already implies the conclusion.*

Proof. The two-offset quotient in Proposition 3.2 recovers the future actions on a common real collar, with the marked physical momenta restored. Thus it suffices to treat equality of $\mathcal{S}(F)$ and $\mathcal{S}(G)$ there. Theorem 4.2 first identifies the curvatures without selecting a local inverse branch. The signed recursion of Theorem 3.4 then identifies every finite contact jet. In particular $h = G - F$ is flat at zero at each physical phase.

The segment between these two actual graph representatives has the same positive quadratic jets. After shrinking the collar it consists of admissible local strictly convex arcs with the fixed stationary polygon. No interpolation of globally closed obstacles is necessary. Lemma 5.1 supplies one $a < 1$ and one common collar for this segment. Now choose a finite integer $m \geq 3$ so large that (5.4) holds. Flatness and Taylor's formula make $\|h\|_{m,R}$ finite. Proposition 5.2 with $g = 0$ forces this norm to vanish. Hence $F = G$ on that collar.

Only finitely many jets are used after a has been chosen. The proof does not infer equality of smooth functions from their Taylor series: it uses equality of their full actions once more, in (5.6). A flat nonzero contact perturbation has a nonzero, possibly flat, action response. $\square$

For connected closed analytic obstacle boundaries, fixed placement and full visitation still give the whole-obstacle conclusion of Theorem F.4.3. For smooth boundaries the new conclusion concerns the visited contact neighborhoods; arbitrary unvisited boundary arcs are not determined by local half-line data. The bounded-holomorphic local inverse of Theorem F.3.6 is also retained: equality of real germs and inversion in a specified analytic Banach norm are different assertions.

5.3. Finite-smoothness control from real densities. The weighted estimate also gives stability of complete real profiles, rather than only finite jets, under a finite-smoothness prior. We state the derivative loss and the range of uniformity explicitly. Fix a marked polygon, a real interval $I_0 = [-R_0, R_0]$, positive numbers $c_- < c_+$, and a bound K_0 . For an integer m let $\mathcal{K}_m$ consist of anchored graph tuples satisfying

$$c_- \leq F_i''(0) \leq c_+, \quad \max_i \|F_i\|_{C^{m+3}(I_0)} \leq K_0. \quad (5.7)$$

We use a sufficiently small common collar of their fixed clear polygon. Curvature remains positive there by the C^3 bound. Choose $a < 1$ larger than all the one-step linear contraction factors for the enlarged quadratic box $[c_-/2, 2c_+]^r$, and then choose $m \geq 3$ satisfying (5.4). This order is fixed by the positive marked class; it need not remain bounded in a weak-hyperbolicity limit.

Lemma 5.4 (Finite-order comparison bounds). *On a sufficiently small common collar $I = [-R, R]$, the following bounds hold with constants depending only on the stipulated class. For $F, G \in \mathcal{K}_m$, if $\eta = \|\mathcal{S}(F) - \mathcal{S}(G)\|_{C^m(I)}$, then*

$$\max_{i, 2 \leq j < m} |F_i^{(j)}(0) - G_i^{(j)}(0)| \leq C\eta. \quad (5.8)$$

If P is an anchored polynomial tuple of degree at most $m - 1$, and the segment $G + tP$ stays in an enlarged bounded positive class, then

$$\|\mathcal{S}(G + P) - \mathcal{S}(G)\|_{C^m(I)} \leq C\|P\|_{C^{m+3}(I_0)}. \quad (5.9)$$

The enlarged class may be fixed before the small perturbation is chosen.

Proof. For degree two, (4.6) gives a uniform bound because the positive action Hessians have a common lower bound. The signed factors depend smoothly on curvature in the enlarged positive box. At every subsequent degree subtract the two inversions (3.12). Their inverse blocks and derivatives are bounded on that box. The remainders $\mathcal{R}_j$ are smooth functions of the finitely many lower graph jets on bounded boxes with positive quadratic entries. Their derivatives there are bounded. Induction, using the observed action-jet differences, proves (5.8). This is a two-point comparison; nonsingularity of an everywhere defined Jacobian is not being used as a substitute for global injectivity or a global Lipschitz bound.

For (5.9), use the weighted stationary equations along $G + tP$. On a common collar their linearized interior operator is a uniformly small perturbation of the positive Jacobi operator. It has a uniform inverse on an exponentially weighted space. Successively differentiating in the endpoint through order m puts the same operator on the highest coordinate derivative. The other terms involve previously bounded derivatives and finitely many graph derivatives. A strict margin in the exponential weight absorbs the fixed polynomial factors in the site index. With some $a < \rho < 1$, this gives

$$|x_{b,j}^t(u)| \leq C\rho^j|u|, \quad \sup_{|u| \leq R} |\partial_u^k x_{b,j}^t(u)| \leq C_k \rho^j \quad (1 \leq k \leq m),$$

uniformly in t . Chord differentiation likewise bounds $\|a_b^t\|_{C^m}$ and $\|v_{b,j}^t\|_{C^m}$ uniformly in b, j, t . The stipulated C^{m+3} graph bound is sufficient for these finitely differentiated equations and weights.

Because $P(0) = P'(0) = 0$, the chain rule gives the useful summable bound

$$\|P_{b+j} \circ x_{b,j}^t\|_{C^m([-R,R])} \leq C\rho^{2j}\|P\|_{C^m(I_0)}.$$

For a derivative involving one derivative of P , its value is $O(|x_{b,j}^t|)$ and its coordinate-derivative factor decays as ρ^j . Terms with at least two derivatives of P have at least two such coordinate factors. The undifferentiated term is quadratic in $x_{b,j}^t$. These observations prove the bound for every order at most m , including the cases where a polynomial derivative is zero. Apply this to the actual envelope with variation P . The initial term has norm at most $C\|P\|_{C^m}$ and the later terms have a summable $C\rho^{2j}\|P\|_{C^m}$ majorant. The terminal term and its fixed derivatives vanish uniformly, as in Lemma 3.3. Thus the parameter derivative of the action has C^m norm at most $C\|P\|_{C^{m+3}(I_0)}$. Integration in t proves the claim. No optimal derivative count or same-regularity differentiability theorem for composition operators is being used. $\square$

Theorem 5.5 (Complete real-profile stability). *For the class (5.7) and m chosen as above, there are a common $R > 0$, $\eta_0 > 0$ and $C < \infty$ such that*

$$\|F - G\|_{C^0([-R,R])} \leq C\|\mathcal{S}(F) - \mathcal{S}(G)\|_{C^m([-R,R])} \quad (5.10)$$

whenever $F, G \in \mathcal{K}_m$ and the norm on the right is at most η_0 . In particular, on compact positive-density classes with fixed separated offsets and residual floors,

$$\|F - G\|_{C^0([-R,R])} \leq C \max_{i,d \in \{d_1,d_2\}} \|f_{i,d}^F - f_{i,d}^G\|_{C^m([-R,R]^2)} \quad (5.11)$$

for sufficiently small right side. These are estimates on realizable pairs of laws. They use a finite real-smoothness prior, not an analytic prior, and do not assert an inverse on arbitrary density tuples.

Proof. Let η be the action norm in the statement and set

$$P_i(u) = \sum_{j=2}^{m-1} \frac{F_i^{(j)}(0) - G_i^{(j)}(0)}{j!} u^j, \quad G^\sharp = G + P.$$

Lemma 5.4 gives $\|P\|_{C^{m+3}(I_0)} \leq C\eta$. For sufficiently small η_0 , both $G + tP$ and the segment between F and $G^\sharp$ remain in a fixed enlarged positive class. Their collars and visit estimates are therefore uniform. The graph tuples $F, G^\sharp$ have identical jets through order $m-1$. By actual finite-jet factorization so do their actions. Taylor's integral remainder consequently gives

$$\|\mathcal{S}(F) - \mathcal{S}(G^\sharp)\|_{m,R} \leq \frac{1}{m!} \|\mathcal{S}(F) - \mathcal{S}(G^\sharp)\|_{C^m([-R,R])} \leq C\eta.$$

The last inequality uses (5.9) and the definition of η . Applying (5.5) to this pair, and then removing the polynomial correction, proves

$$\|F - G\|_{C^0} \leq R^m \|F - G^\sharp\|_{m,R} + \|P\|_{C^0} \leq C\eta.$$

All collars were fixed from the enlarged class before this comparison. Finally, Proposition 3.2 converts the stated C^m density error to C^m action error on the positive square, proving (5.11). $\square$

The theorem also proves exact contact identification on each of these finite-smoothness classes. Its quantitative conclusion is a C^0 profile estimate from a C^m law error with a C^{m+3} graph bound. It is not an analytic-norm estimate, a derivative-free statistical rate, or a theorem uniform in the period. If two realizable candidates fit common finite-flight density records to accuracy ε in this same C^m norm, the relative theorem at order m gives

$$\|F - G\|_{C^0([-R,R])} \leq C(\tau^N + \varepsilon)$$

whenever the combined error is small and the forward constants are uniform on the class. This uses the triangle inequality through the common records. It does not manufacture density derivative bounds from total variation or assign a preparation budget. The curvature stopping error in Proposition F.4.4 remains a separate, explicitly propagated computational error.

5.4. A smooth physical distinction beyond all contact jets.

Proposition 5.6 (Flat contacts distinguished by boundary laws). *Let a smooth strictly dispersing table have a clear nongrazing marked periodic polygon with distinct contact points. There is a small smooth family of actual tables, with the same polygon and free cell area, whose complete contact Taylor jets at every marked contact agree, but whose two-offset boundary laws do not agree on any sufficiently small common contact square. Every future action also has the same Taylor jets throughout the family. Thus a formal all-order contact record and the complete boundary-law record have different geometric information even in the fixed marked physical class.*

Proof. Choose one contact, called b , and a smooth nonnegative cutoff χ supported in its graph chart and equal to one near zero. Replace its graph by

$$F_b^\varepsilon(u) = F_b^0(u) + \varepsilon\psi(u), \quad \psi(u) = \begin{cases} \chi(u)e^{-1/u^2}, & u \neq 0, \\ 0, & u = 0. \end{cases}$$

All other contact graphs are unchanged. The difference is flat but nonzero on every punctured neighborhood. For small ε , strict convexity, embeddedness, disjointness and the clearance margins persist. The fixed contact points, tangent frames and curvatures preserve the original periodic polygon exactly.

To keep the free area fixed, choose a smooth normal bump on a boundary arc disjoint from every contact collar, with nonzero first area variation. Such an arc exists after the finitely many collars are shrunk. For a positively oriented obstacle boundary $R(s)$, its area is $\frac{1}{2} \int R \times R' ds$; differentiation and integration by parts give first variation $\int \delta R \times R' ds$. A nonzero nonnegative outward normal bump therefore has nonzero area derivative. The ordinary implicit function theorem chooses its coefficient smoothly in ε so that the obstacle area, hence the free cell area, is unchanged. The correction is outside all the relevant charts and leaves their stationary half-lines unchanged. This constructs actual closed obstacles, rather than only formal local coefficients.

All graph jets remain fixed. The actual finite-jet factorization therefore fixes every action jet as well. The exact linear reference twists, flight lengths, incidence angles and multipliers are also unchanged. Nevertheless the envelope on the perturbed contact detects the difference. On a smaller collar, (5.2) and positivity of ψ give, for $\varepsilon > 0$ and $u \neq 0$,

$$S_b^{-,\varepsilon}(u) - S_b^{-,0}(u) \geq \frac{\varepsilon}{2} e^{-1/u^2} > 0.$$

The area-correcting bump contributes nothing to this local action. More precisely, uniformly for a sufficiently small parameter range,

$$S_b^{-,\varepsilon}(u) - S_b^{-,0}(u) = \varepsilon e^{-1/u^2} \left\{ 1 + O(|u|) + O\left(e^{-(a^{-2}-1)/u^2}\right) \right\}. \quad (5.12)$$

Indeed $\alpha_b^t(u) = 1 + O(|u|)$, while each later perturbed visit has $|x_{b,j}^t(u)| \leq a^j |u|$. Dividing its flat value by e^{-1/u^2} and using $a^{-2j} - 1 \geq j(a^{-2} - 1)$ bounds the entire tail by the convergent geometric series with ratio $e^{-(a^{-2}-1)/u^2}$. Visits to unperturbed phases contribute zero.

Equality of the two-offset laws would imply equality of these actions by the anchored quotient (3.6), a contradiction. The claim is not equality of the complete fixed-window count or of the normalized density Taylor coefficients: their normalizers can change. It is equality of all formal contact and action data, with the reference geometry and area fixed, in the presence of a nonzero

physical-law signal. Nor is it a comparison of full marked length spectra, which are not the observations used here. $\square$

The flat family explains why the complete-law hypothesis in Theorem 5.3 is substantive. It is not a formal consequence of all-order contact separation. Exact relative normalization first retains the real action functions in the physical law; the contracted visits then identify those functions with actual boundary profiles, including their flat parts. The finite-smoothness estimate records the derivative information needed for stable use of that same mechanism.

5.5. A phase-dependent inverse for actual functions. The scalar tail estimate above can be organized by the phases visited by a half-line. This gives an inverse criterion that retains heterogeneous contraction and separates it from the conditioning of finite jets. The criterion applies to actual functions and to an integrated nonlinear envelope; it does not require an analytic power series.

Proposition 5.7 (Matrix domination of a contracted envelope). *Let h_b, g_b be continuous functions on $[-R, R]$, $0 \leq b < r$, vanishing at zero and with finite weighted order- m norms. Suppose*

$$g_b(u) = \alpha_b(u)h_b(u) + \int_0^1 \sum_{j \geq 1} v_{b,j}^t(u) h_{c(b,j)}(\psi_{b,j}^t(u)) dt, \quad (5.13)$$

where $\alpha_b \geq \alpha_* > 0$, $|v_{b,j}^t| \leq V_{b,j}$ and $|\psi_{b,j}^t(u)| \leq \gamma_{b,j}|u|$ with $0 \leq \gamma_{b,j} \leq 1$. Define the nonnegative matrix

$$K_m(b, c) = \alpha_*^{-1} \sum_{j: c(b,j)=c} V_{b,j} \gamma_{b,j}^m. \quad (5.14)$$

Assume its entries are finite and its spectral radius is less than one. Writing

$$H_b = \sup_{0 < |u| \leq R} \frac{|h_b(u)|}{|u|^m}, \quad G_b = \sup_{0 < |u| \leq R} \frac{|g_b(u)|}{|u|^m},$$

one has the componentwise estimate

$$H \leq \alpha_*^{-1} (I - K_m)^{-1} G. \quad (5.15)$$

In particular, if $w_b > 0$ and $K_m w \leq \theta w$ for some $\theta < 1$, then

$$\|h\|_{m,R,w} \leq \frac{\alpha_*^{-1}}{1-\theta} \|g\|_{m,R,w}, \quad \|h\|_{m,R,w} := \max_b \sup_{0 < |u| \leq R} \frac{|h_b(u)|}{w_b |u|^m}. \quad (5.16)$$

Such a weight exists whenever the stated spectral-radius condition holds.

Proof. The summability in (5.14) is an absolute majorant for the envelope on the weighted space. Dividing (5.13) by $\alpha_b(u)|u|^m$ and taking suprema yields $H \leq \alpha_*^{-1} G + K_m H$. Iterate this inequality n times. The finite matrix K_m^n tends to zero and $\sum_{j \geq 0} K_m^j = (I - K_m)^{-1}$, whose entries are nonnegative. Taking the limit proves (5.15). For a weight satisfying $K_m w \leq \theta w$, divide the componentwise inequality by w_b and take a maximum to prove (5.16). Conversely choose any $\theta \in (\rho(K_m), 1)$ and set $w = (I - K_m/\theta)^{-1} \mathbf{1}$. Then $w \geq \mathbf{1}$ and $K_m w = \theta(w - \mathbf{1}) \leq \theta w$. All statements concern a fixed integrated envelope. Its coefficients may depend on both candidates; no observation-only iteration is asserted. $\square$

Corollary 5.8 (Phase-weighted smooth separation). *For the graph interpolation in Proposition 5.2, suppose the one-step maps satisfy, on a common collar and for all interpolation parameters,*

$$|(X_i^t)'(u)| \leq \hat{a}_i < 1, \quad 0 < \hat{a}_i, \quad \bar{a} = \left(\prod_{i=0}^{r-1} \hat{a}_i \right)^{1/r}.$$

For any integer $m \geq 3$ such that $\bar{a}^m < 1/7$, two graph tuples with equal jets through degree $m - 1$ satisfy

$$\|G - F\|_{m,R,w} \leq \frac{2}{1 - \vartheta_m} \|\mathcal{S}(G) - \mathcal{S}(F)\|_{m,R,w}, \quad \vartheta_m = \frac{6\bar{a}^m}{1 - \bar{a}^m}, \quad (5.17)$$

where $w_0 = 1$ and

$$w_b = \left(\prod_{i=0}^{b-1} \frac{\bar{a}}{\hat{a}_i} \right)^m \quad (1 \leq b < r). \quad (5.18)$$

In the unweighted norm the right-side constant is multiplied by $\kappa(w) := \max_b w_b / \min_b w_b$.

Proof. The mean value theorem and the constant-one composition identity in Lemma 5.1 give

$$|x_{b,j}^t(u)| \leq \left(\prod_{\ell=0}^{j-1} \hat{a}_{b+\ell} \right) |u|.$$

Use $\alpha_* = 1/2$, $V_{b,j} = 3$ and $c(b, j) = b + j \pmod{r}$ in Proposition 5.7. If $(A_m z)_b = \hat{a}_b^m z_{b+1}$, its domination matrix is

$$K_m = 6 \sum_{j \geq 1} A_m^j = 6A_m(I - A_m)^{-1}.$$

The positive weight (5.18) obeys $A_m w = \bar{a}^m w$, including the closing phase, because $\bar{a}^r = \prod_i \hat{a}_i$. It follows that $K_m w = \vartheta_m w$. In fact conjugation by $\text{diag}(w)$ changes A_m into $\bar{a}^m$ times the cyclic permutation matrix, so its nonnegative geometric sum has spectral radius ϑ_m . The hypothesis is exactly $\vartheta_m < 1$. The preceding proposition proves (5.17). The norm inequalities

$$\|h\|_{m,R} \leq (\max_b w_b) \|h\|_{m,R,w}, \quad \|g\|_{m,R,w} \leq (\min_b w_b)^{-1} \|g\|_{m,R}$$

prove the unweighted estimate. The actual envelope, its terminal-limit justification and the finite-jet factorization are those already proved; only the estimate of its tail has changed. $\square$

Corollary 5.9 (Finite-smoothness inverse with phase bounds). *Fix a bounded positive marked class and an enlarged class for the polynomial alignments of Lemma 5.4. Suppose the latter has common one-step bounds $\hat{a}_i$ as above. Choose $m \geq 3$ with $\bar{a}^m < 1/7$ and retain the C^{m+3} graph prior and the positive density, gate and offset floors. Then the conclusions (5.10) and (5.11) hold at this order m on a smaller common collar, for sufficiently small data discrepancy.*

Proof. The finite-jet comparison lemma holds at every stipulated finite order: its proof uses the global quadratic inverse, the signed finite blocks and the forward polynomial variation estimate, not the scalar tail threshold. Set $\eta = \|\mathcal{S}(F) - \mathcal{S}(G)\|_{C^m}$ and form its polynomial correction P and $G^\sharp = G + P$ exactly as in Theorem 5.5. Then $\|P\|_{C^{m+3}} \leq C\eta$. For small enough η , both required interpolations remain in the fixed enlarged positive class. The actions of F and $G^\sharp$ have equal jets through order $m - 1$, and Taylor's formula and the forward variation bound give

$$\|\mathcal{S}(F) - \mathcal{S}(G^\sharp)\|_{m,R,w} \leq \frac{C\eta}{\min_b w_b}.$$

Apply (5.17), multiply by $R^m \max_b w_b$, and add $\|P\|_{C^0}$. This proves the action estimate with a finite class-dependent constant. The two-offset quotient on the same positive gate gives the density estimate. No curvature bound is inferred from the resulting C^0 graph error; curvatures were controlled in the preceding jet alignment. $\square$

The choice $\hat{a}_i = a$ recovers the previous sufficient condition and constant. On a heterogeneous class the geometric-mean criterion may allow a smaller order than $6(\max_i \hat{a}_i)^m / [1 - (\max_i \hat{a}_i)^m] < 1$. The price is the explicit weight condition number, as well as the unchanged finite-jet constants. Thus this refinement improves a sufficient inverse criterion without asserting that the selected

order is optimal or that the unweighted inverse is uniformly better. It also illustrates the reusable content of the smooth argument: invert the finite jets, then dominate the remaining actual-function envelope by a nonnegative visit matrix.

6. LOCAL DETERMINATION AND EXTERIOR SCATTERING DATA

The endpoint law is constructed from actual finite trajectories. We now make precise its dependence on a neighborhood of the marked itinerary. This also fixes the comparison with the global scattering observation in Section 1.4.

6.1. The local observation record. Fix the polygon, its marked frames, a sufficiently small common graph interval I , and the two offsets of Theorem 2.5. Shrink I once, if necessary, so that the stationary bridges for all returning $N \geq N_0$ lie in prescribed contact collars with compact closure. There are finitely many such collars modulo the lattice. Their successive chords, together with the short initial and terminal free segments allowed by the time-split construction, lie in a fixed neighborhood U of the itinerary modulo the lattice. We require the same clearance and free-segment margins as in that theorem. The neighborhood includes the reflecting collars; away from them its relevant trajectory tubes are obstacle-free.

A *local completion* consists of closed obstacles agreeing with the prescribed contact graphs on these collars and satisfying these margins on U ; the boundary away from U may vary. The lattice and marks are fixed. This definition concerns actual tables, not independently assigned Taylor coefficients. For a chosen phase b , returning length N and offset d , a reset preparation is drawn from the normalized Liouville measure of that table. Record the endpoint mark (u, v) on success of the stipulated itinerary event, and a single symbol $\dagger$ otherwise. Write $\mu_{N,b,d}$ for the success subprobability measure on the gate Q_+ , $p_{N,b,d} = \mu_{N,b,d}(Q_+)$, and

$$P_{N,b,d} = (1 - p_{N,b,d})\delta_{\dagger} + \mu_{N,b,d}. \quad (6.1)$$

The conditional endpoint density is that of $\mu_{N,b,d}/p_{N,b,d}$. The symbol $\dagger$ contains no trajectory trace or additional data from failed preparations. Both the resetting and this recording convention are essential to the statement below.

Proposition 6.1 (Exact locality of the charged record). *Let Ω and $\tilde{\Omega}$ be two local completions with the same contact graphs on the prescribed collars, and free areas $\mathcal{A}$ and $\tilde{\mathcal{A}}$ in the common cell. Then, for every phase, both selected offsets, and every returning $N \geq N_0$,*

$$\tilde{\mu}_{N,b,d} = \frac{\mathcal{A}}{\tilde{\mathcal{A}}} \mu_{N,b,d}. \quad (6.2)$$

In particular the conditional finite-flight densities and their limiting laws coincide. If $\mathcal{A} = \tilde{\mathcal{A}}$, the complete records (6.1) coincide, and so do all finite products of these independently reset experiments. The assertion is exact, not an asymptotic estimate as $N \rightarrow \infty$.

Proof. The single-edge length functions between successive collars are the same in the two tables. The stationary equations and their boundary conditions are consequently identical. The weighted uniqueness in Theorem 2.3, on the chosen common collars, identifies their finite stationary bridges. They remain physical in each table by the common clearance margins. Thus their stationary lengths $W_{N,b}$, gauged actions $E_{N,b}$, mixed twists and reference normalizations $D_{N,b}$ agree as functions of the endpoints. Equality holds throughout the endpoint gate, not just to finite order at its origin.

Put $\beta_{N,b} = (-\partial_{uv}W_{N,b})/D_{N,b}$ and $r_{N,b,d} = d - E_{N,b} + p_b u - p_b v$. The exact time-split computation of Theorem 2.5 gives

$$\frac{d\mu_{N,b,d}}{du dv} = \frac{D_{N,b}}{2\pi\mathcal{A}} \beta_{N,b}(u, v) \{r_{N,b,d}(u, v)\}_+. \quad (6.3)$$

Indeed the allowed initial free time ranges from 0 to $r_{N,b,d}$; the free-segment margins ensure that neither an earlier nor an additional reflection truncates this interval. Every successful trajectory is accounted for by this parametrization because the event specifies all intermediate collisions in the collars and the last-collision rule. It does not include other itineraries elsewhere in the table. All factors in (6.3) except the total free area are common to the two completions. This proves (6.2). Integrating gives the same factor for $p_{N,b,d}$, so normalization cancels it. The positive success floor ensures that the conditional densities are defined. Their limits agree by the relative law. At equal area the subprobabilities and hence the failure masses agree; independence gives the final product assertion. $\square$

Remark 6.2. If settings are selected from the permitted experiments by the same measurable rule based on earlier recorded outcomes and independent auxiliary randomization, equality also holds for every fixed finite number of preparations. Conditional on the history the next setting has the same distribution and its record kernel is the same; induction proves the assertion. This does not extend it to an unreset trajectory, to recording a rejected ray's path, or to a table-dependent error kernel. A common recording kernel applied to the exact records preserves equality.

6.2. Complete contact information without remote completion. Two marked tables will be called *contact equivalent* when their graph functions agree on some common neighborhood of every contact. Use a germ of the local law experiment: endpoints may be restricted to a smaller common positive-residual gate, with its normalization specified anew. This convention is necessary when only contact germs, rather than fixed-radius graphs, are given. In particular, restricting a density without renormalizing its original larger gate is a different operation.

Corollary 6.3 (Local completion and the determined object). *On the marked smooth class of Theorem 1.1, equality of the two-offset limiting law germs is equivalent to contact equivalence. Here equality of laws is tested on a sufficiently small common gate at every phase. In the equal-area subclass, contact equivalence also gives equality of all charged finite-flight records on sufficiently small common gates.*

There are nonconstant smooth one-parameter families of closed strictly convex obstacles in a fixed periodic cell, with the same free area and marked polygon, for which these charged records are identical although the obstacle boundaries away from the collars differ. These families can be chosen so that their finite compact obstacle unions in the plane have different almost-everywhere exterior travelling-time data and different almost-everywhere scattering length spectra.

Proof. Equality of the limiting laws gives equality of the anchored action functions by Proposition 3.2. The global quadratic inverse, signed finite-jet inversion, and actual smooth inverse of Theorem 5.3 then identify the graph functions on a common smaller collar. These steps use actual realizable law tuples. Conversely, if the contact germs agree, choose compact collars inside their common domains. The weighted finite and half-line constructions permit a sufficiently small common endpoint gate for which all stationary visits stay inside these collars. Both tables have positive clearance along the fixed polygon, so the short endpoint segments also have common margins. The two offsets are chosen below these common free-flight bounds, as required by the experiment. Proposition 6.1 applies. It proves the limiting assertion, and the finite-record assertion when the areas agree. This is an equivalence of germs of experiments; it asserts no equality on a larger gate outside the common graph domains.

For an actual family take the three-obstacle realization of Proposition 2.6. Choose U sufficiently small around its contact collars and flight tubes. On an obstacle boundary choose two disjoint open arcs with closures outside U . Such arcs exist on the side facing away from the interior contact polygon in this explicit configuration. Let ψ_1, ψ_2 be nonzero smooth nonnegative normal bumps supported in these arcs. For small real parameters (s, t) deform the boundary normally by

$s\psi_1 + t\psi_2$, leaving all other boundary points unchanged. Disjoint compact embedded boundaries stay disjoint and strictly convex under sufficiently small C^2 perturbations; the lattice translates remain separated as well. The positive distance between the bump supports and the closed trajectory tubes preserves all the required local margins.

Let $V(s, t)$ be the total obstacle area in the cell. Normal variation of area gives

$$\partial_t V(0, 0) = \int \psi_2 d\sigma > 0.$$

For completeness this derivative follows by using boundary normal coordinates: a strip of signed width $t\psi_2(\sigma)$ has area $t \int \psi_2 d\sigma + O(t^2)$, with a uniform remainder on the compact support. The area is a smooth function of the two parameters. The implicit function theorem gives a smooth $t = t(s)$ near zero, with $t(0) = 0$ and $V(s, t(s)) = V(0, 0)$. The family is nonconstant, since on the first support its normal displacement is exactly $s\psi_1$ and the second bump has disjoint support. All contact graphs and marks are unchanged. The free area is constant, so Proposition 6.1 proves equality of every indicated finite record. The observation samples are still drawn from the full table, not from a law postulated only on U .

Regard the same three obstacles, without their lattice copies, as a compact union K_s in $\mathbb{R}^2$. All K_s lie in one fixed enclosing circle and satisfy the strictly convex C^3 hypotheses of Noakes–Stoyanov [5, Theorem 1.1]. For $s \neq s'$ sufficiently small, $K_s \neq K_{s'}$ by the disjoint-support construction. That theorem implies that their travelling-time families are not equal on a full-measure set of incoming positions and directions. Its separate scattering-length conclusion gives the analogous distinction for incoming/outgoing direction pairs. The periodic records remain identical by the preceding paragraph. This proves the asserted comparison on an explicitly common class of geometric realizations. $\square$

The two parts of this comparison serve different purposes. In the local experiment, the inverse determines the actual visited contact germs, including flat changes, whereas the construction varies only unvisited boundary pieces and leaves even the finite charged record fixed. The exterior observation distinguishes those pieces by a different data family. Hence equality of the local record cannot simply be substituted for the equality hypothesis in a global lens theorem. There is no claim that local laws distinguish equal full marked length spectra, nor that a known whole obstacle and the supplied marks could not generate its local laws. The comparison sharpens the local determination statement without weakening it and without identifying the two observation models.

Part 2. Consequences for the observation record

7. ONE UNCALIBRATED LAW AT AN OBLIQUE CONTACT

The two-offset observation separates unequal actions without using their first derivatives. At an oblique contact those derivatives are already fixed by the geometric marks. They remove both the multiplicative ambiguity of a rank-one factorization and the unknown time scale. This gives a second observation theorem for the same smooth inverse, without changing the two-offset theorem at normal incidence.

7.1. An anchored factorization and its scale. Let $I = [-R, R]$. Consider a positive C^M function on a neighborhood of I^2 , $M \geq 2$, of the form

$$f(u, v) = Z^{-1} B_-(u) B_+(v) \{d - A(u) - C(v)\}, \quad (7.1)$$

where $d > 0$, $Z > 0$, $B_{\pm} > 0$, $B_{\pm}(0) = 1$, and

$$A(0) = C(0) = 0, \quad A'(0) = -p, \quad C'(0) = p, \quad p \neq 0. \quad (7.2)$$

The signed number p is known; d , both amplitudes and Z are unknown. Assume $d - A - C > 0$ on I^2 . No equality of A and C is imposed. Multiplication of f by an unknown positive constant is permitted. Define

$$\mathcal{R}_f(u, v) = \frac{f(u, v)f(0, 0)}{f(u, 0)f(0, v)}, \quad \kappa_f = \partial_u \partial_v \mathcal{R}_f(0, 0). \quad (7.3)$$

For a continuous positive density, its law determines its values on the square, and thus these expressions whenever the indicated derivatives exist. Finite samples are treated below, not identified with pointwise density observations.

Proposition 7.1 (Identification of the unknown offset). *For (7.1)–(7.2),*

$$\mathcal{R}_f = 1 - UV, \quad U = \frac{A}{d - A}, \quad V = \frac{C}{d - C}, \quad \kappa_f = \frac{p^2}{d^2} > 0, \quad d = \frac{|p|}{\sqrt{\kappa_f}}. \quad (7.4)$$

Consequently the offset is determined by the single conditional law and the marked momentum, without a count or an amplitude model.

Proof. The axis residuals $d - A(u)$ and $d - C(v)$ are positive. Cancellation of Z and the two amplitudes gives

$$\mathcal{R}_f(u, v) = \frac{d\{d - A(u) - C(v)\}}{(d - A(u))(d - C(v))} = 1 - \frac{A(u)C(v)}{(d - A(u))(d - C(v))}.$$

Since $U(0) = V(0) = 0$, $U'(0) = -p/d$ and $V'(0) = p/d$, $\kappa_f = -U'(0)V'(0) = p^2/d^2$. The positive choice of d fixes the square-root branch. Equivalently $\kappa_f = \partial_{uv} \log f(0, 0)$, because $\mathcal{R}_f$ equals one on both axes. This identity is insensitive to the unknown normalization or separate endpoint factors. $\square$

The clock before the relative limit. For the actual finite-flight density write $f_N = Z_N^{-1} \beta_N r_N$, where $r_N = d - E_N + pu - pv$. The exact finite-flight counterpart of (7.4) is

$$\partial_{uv} \log f_N(0, 0) = \frac{p^2}{d^2} + \frac{D_N}{d} + \partial_{uv} \log \beta_N(0, 0). \quad (7.5)$$

Indeed $r_N(0, 0) = d$, $r_{N,u}(0, 0) = p$, $r_{N,v}(0, 0) = -p$ and $r_{N,uv}(0, 0) = D_N$. Differentiating $\log r_N$ proves the formula. The normalization is constant, and the remaining term is the indicated derivative of the amplitude. This identity was pointed out in [6]. The mixed derivative of the anchored finite-density cross ratio has the same value, because that cross ratio equals one on both axes.

The last term in (7.5) has no asserted sign. The relative C^2 theorem and positive floors show that its sum with D_N/d is $O(e^{-\omega N})$ on the bounded class; one may decrease $\omega > 0$ to a common strict exponential margin. In particular, if $\kappa_* = p_*^2/d_+^2$ and the error is at most $\kappa_*/2$, the finite mixed derivative is positive and the direct plug-in clock differs from d by at most $Ce^{-\omega N}$. This follows from the mean value bound for $x \mapsto |p|x^{-1/2}$ on $[\kappa_*/2, \infty)$. It is a biased finite-flight approximation, not the exact identity (7.4). The forward term in (7.12) includes this bias.

The rank-one identity alone would determine U, V only up to $(U, V) \mapsto (\lambda U, \lambda^{-1} V)$. More precisely, if neither factor is identically zero and two pairs give the same product, evaluate at a point where the second factor is nonzero to identify the first factor up to a scalar, and then identify the second. The specified first derivatives in (7.2), after (7.4) has fixed d , force $\lambda = 1$. The following formulas implement that argument without differentiating a whole recovered function one extra time.

Choose one fixed $a \in I \setminus \{0\}$ sufficiently close to zero that $A(a)C(a) \neq 0$. Such an a exists by (7.2). Using the recovered d , put

$$v_a(f) = \frac{d}{p} \partial_u \mathcal{R}_f(0, a), \quad u_a(f) = -\frac{d}{p} \partial_v \mathcal{R}_f(a, 0), \quad (7.6)$$

$$U_f(u) = \frac{1 - \mathcal{R}_f(u, a)}{v_a(f)}, \quad V_f(v) = \frac{1 - \mathcal{R}_f(a, v)}{u_a(f)}. \quad (7.7)$$

The quantities u_a, v_a are signed scalars. They are not positive square roots and must not be replaced by their absolute values.

Theorem 7.2 (One-law separation with an unreported offset). *The single function f in (7.1) and the known $p \neq 0$ determine d, A, C, B_-, B_+ uniquely on I , with*

$$A_f(u) = \frac{dU_f(u)}{1 + U_f(u)}, \quad C_f(v) = \frac{dV_f(v)}{1 + V_f(v)}, \quad (7.8)$$

$$B_{-,f}(u) = \frac{f(u, 0)}{f(0, 0)}(1 + U_f(u)), \quad B_{+,f}(v) = \frac{f(0, v)}{f(0, 0)}(1 + V_f(v)). \quad (7.9)$$

For $M \geq 2$ this inverse is locally Lipschitz from $C^M(I^2)$ to $\mathbb{R} \times C^M(I)^4$ on classes with uniform bounds for the C^M norms and positive lower bounds for $f, \kappa_f, |u_a(f)|, |v_a(f)|, 1 + U_f$, and $1 + V_f$. The formulas extend to an open neighborhood of each such bounded class, even when a perturbed function does not have representation (7.1).

Proof. Differentiating $1 - UV$ at $(0, a)$ and $(a, 0)$ gives $v_a(f) = V(a)$ and $u_a(f) = U(a)$. Thus (7.7) recovers the actual factors on all of I , including their zeros. Solving the definitions of U, V gives (7.8). Both final denominators are positive, since $1 + U = d/(d - A)$ and $1 + V = d/(d - C)$. The two axis identities in (7.1) give (7.9). This also proves uniqueness: any other representation satisfying the same slopes gives the same offset, factors, actions and normalized amplitudes.

For stability, restriction to a fixed slice is bounded in C^M . Products and reciprocal functions are locally Lipschitz on bounded C^M sets with positive denominator floors. Apply these facts first to $\mathcal{R}_f$. Evaluation of its mixed derivative at the single point $(0, 0)$ is a bounded scalar functional when $M \geq 2$. The function $x \mapsto |p|x^{-1/2}$ is Lipschitz on an interval with a positive lower endpoint, so d is locally Lipschitz. The two first-derivative point evaluations in (7.6) are bounded scalar functionals as well. The stipulated nonzero anchor bounds permit division in (7.7); fixed slices give C^M functions, with no loss of one derivative. Finally apply the product and reciprocal bounds to (7.8)–(7.9). For small enough perturbations all floors persist, possibly halved. No assertion that the extended inverse produces an actual billiard is made. $\square$

Uniform anchors follow from geometric priors rather than an assumed conditioning bound on a measured inverse. Suppose $|p| \geq p_* > 0$, $d_- \leq d \leq d_+$, and the C^2 norms of A, C are bounded by H_2 . Choose a fixed $a > 0$ with $H_2 a \leq p_*$, inside the common collar. Taylor's formula gives $|A(a)|, |C(a)| \geq p_* a/2$. If the positive axis residuals are bounded above by D_* , then

$$|u_a|, |v_a| \geq \frac{p_* a}{2D_*}, \quad \kappa_f \geq \frac{p_*^2}{d_+^2}, \quad 1 + U, 1 + V \geq \frac{d_-}{D_*}.$$

The physical relative law gives uniform amplitude and normalized-density bounds on a common positive square. These observations supply all the floors used in Theorem 7.2. Constants need not remain bounded as p_* or a tends to zero.

7.2. Smooth contact determination. We return to the fixed marked polygon of Section 3. Call it *oblique* if $p_b \neq 0$ at every physical phase. At each phase select one positive offset d_b , fixed over the returning lengths and the independent preparations, but not included in the reconstruction data. The offsets may differ between phases. They lie in a stipulated positive physical window; the contact gate and all event tags retain their exact meanings. Thus an unreported offset is a fixed nuisance parameter, not a random offset, an incorrectly recorded success tag, or an uncertain endpoint gate. No new clock-dependent observation is being supplied.

Theorem 7.3 (Smooth rigidity from one uncalibrated law per phase). *For an oblique clear nongrazing marked polygon with distinct physical contacts and positive-curvature C^∞ graphs, the collection*

$$(f_{b,d_b} : b = 0, \dots, r-1)$$

of single-offset limiting endpoint laws determines all the offsets d_b and the complete smooth reflecting contact germs. Neither the offsets, curvature, flux amplitudes, nor acceptance probabilities are supplied. No symmetry, analyticity or closeness between the candidates is required.

On a bounded positive marked class as in Theorem 5.5, choose $m \geq 3$ so that

$$\frac{6a^m}{1-a^m} < 1$$

on the enlarged positive quadratic class. With uniform obliquity, offsets in $[d_-, d_+]$, and common positive-residual margins, there are $R, C, \eta_0 > 0$ such that

$$\|F - G\|_{C^0([-R, R])} + \|d - \tilde{d}\|_{\ell^\infty} \leq C \max_b \|f_{b,d_b}^F - f_{b,d_b}^G\|_{C^m([-R, R]^2)} \quad (7.10)$$

when the maximum on the right is at most η_0 . These are realizable law pairs; their normalizing constants need not coincide.

Proof. Lemma 3.1 supplies the actual physical law (7.1), with $A_b = S_b^- - p_b u$ and $C_b = S_b^+ + p_b v$. The half-line actions are anchored, so their physical slopes are exactly (7.2). Theorem 7.2 therefore recovers d_b and both actions at every phase. Add the known $p_b u$ to A_b to obtain the future action map $\mathcal{S}(F)$. The action version of Theorem 5.3 identifies the actual contact germs. In particular the last step still uses the contracted-visit envelope on flat differences, not just equality of Taylor series.

On the bounded class choose the common collar and fixed anchor using the preceding uniform bounds. The inverse in Theorem 7.2 gives C^m control of both actions and scalar control of the offsets from the displayed law distance. Apply the action estimate (5.10); its polynomial alignment has already been proved before the weighted norm is used. This gives (7.10). There is no extra derivative loss from clock recovery, since $m \geq 3$ and the point functional in (7.4) has order only two. $\square$

Lemma 7.4 (Normal incidence and distinct physical contacts). *A clear nongrazing periodic billiard polygon with $r \geq 3$ distinct physical contacts is oblique. If a periodic orbit with distinct physical contacts has a normal reflection, it is the ordinary self-retracing two-contact orbit, with zero closing translation.*

Proof. At a normal reflection the outgoing ray is the incoming ray traversed backwards. Its first collision is therefore exactly the preceding lifted contact: the incoming open flight was clear. Consequently the preceding and following physical contacts coincide. For $r \geq 3$ these are different cyclic positions, contradicting distinctness. The only remaining case is $r = 2$. Here the equality of the preceding and following lifted, not merely physical, contact forces zero closing translation. The two flights traverse the same segment in opposite directions; periodicity forces normal reflection at the other endpoint as well. This proves the assertion without a genericity assumption. $\square$

Thus the one-law theorem applies to every polygon with at least three distinct physical contacts in the main theorem, not just a generic subset of those polygons. At normal incidence $\kappa_f = 0$, and the new clock formula is not defined. This is a boundary of this argument, not an impossibility claim about all observations at a normal contact. The general two-calibrated-offset theorem is unchanged. For a self-retracing channel at one *known* offset, the separate symmetric factorization of Theorem F.23.1 also remains available. It uses a nonzero action anchor instead of a marked nonzero first derivative and does not identify an unknown offset by (7.4).

The equal-area flat family of Proposition 5.6 can be based on the actual three-contact configuration of Proposition 2.6. Its two distinct tables cannot have equal single-offset limiting laws at every phase, even if the nuisance offsets are allowed to differ: equality would first force equal offsets and then equal smooth germs by Theorem 7.3. Flat contact information therefore survives the removal of the second offset and its calibration. Conversely, identical contact collars with identical offsets still give the exact local-completion invariance of Proposition 6.1. The new theorem changes the local observation, not its target to unvisited boundary arcs.

7.3. Finite preparations with joint offset recovery. Let $\mathcal{U}$ be a nonempty class of pairs (T, d) consisting of an actual marked table and one fixed offset per phase. Impose the bounded positive contact and forward derivative hypotheses of Section A, replacing the two known offsets by $d_b \in [d_-, d_+]$ and adding the obliquity floor $|p_b| \geq p_*$. All positive-residual, clearance and free-segment margins hold uniformly over that interval. The finite higher graph bounds are those needed for order $m + s$, where m is the same weighted inverse order and $s \geq 1$. The free area has a fixed upper bound.

There are now $J = r$ groups. A preparation in group b uses the actual fixed d_b ; it records the same exact itinerary and Q_+ acceptance tag as before, followed on success by an endpoint pair with bounded post-acceptance error δ . Densities normalized on Q_+ are restricted to $Q \in Q_+$ without renormalization. No additional clock reading, calibration trial or endpoint height is used. Preparations are independent resets, not successive pieces of one orbit.

Theorem 7.5 (Charged joint recovery of profiles and offsets). *On $\mathcal{U}$ there is a measurable estimator $(\hat{T}, \hat{d}) \in \mathcal{U}$ with the following property. Let B preparations be made in each of the $J = r$ groups, and set*

$$\alpha = \frac{s}{2(m+s)+2}, \quad \beta = \frac{\alpha\omega}{\omega + \alpha\Gamma}, \quad \gamma = \frac{s}{m+s+3}, \quad L_B = \log(C_*JB/\zeta),$$

where ω, Γ, c_0, N_0 are uniform forward constants. Choose

$$N_B = P \left\lfloor \frac{\alpha \log(B/L_B)}{P(\omega + \alpha\Gamma)} \right\rfloor, \quad n_B = \lfloor Bc_0e^{-\Gamma N_B}/2 \rfloor.$$

For $0 < \zeta < 1/2$, assume $N_B \geq N_0$, $n_B \geq 2$ and $Bc_0e^{-\Gamma N_B} \geq 8 \log(J/\zeta)$. Impose also the upper bandwidth and small-error conditions in the proof below. Then

$$\sup_{(T,d) \in \mathcal{U}} \mathbb{P}_{T,d} \left(\|F^{\hat{T}} - F^T\|_{C^0([-R,R])} + \|\hat{d} - d\|_{\ell^\infty} > C\{(L_B/B)^\beta + \delta^\gamma\} \right) \leq 2\zeta. \quad (7.11)$$

All rB preparations, including failures, are charged. The estimator is an existence construction on the realizable image and the rate is a sufficient upper bound, not an efficiency or minimax assertion.

Proof. First check the forward hypotheses with the nuisance parameter included. The offset enters the physical numerator affinely. Its compact range, the positive residual integral and the differentiated relative theorem therefore give, uniformly in (T, d) ,

$$\|f_{N,b,d_b}\|_{C^{m+s}(Q_+)} + \|f_{b,d_b}\|_{C^{m+s}(Q_+)} \leq H, \quad \|f_{N,b,d_b} - f_{b,d_b}\|_{C^m(Q_+)} \leq C_0e^{-\omega N}.$$

The same exact reference twist and area upper bound as in (A.3) give success probability at least $c_0 e^{-\Gamma N}$. Knowledge of the actual value of d_b is not used by any of these uniform bounds. The limiting densities also have a common positive floor, so Theorem 7.3 applies.

For n accepted endpoints in each group, use the kernel estimator of Lemma A.1. With $L_n = \log(C_1 J n / \zeta)$ and $n^{-1} \leq h \leq h_0$, its error relative to the limiting density tuple, in the product $C^m(Q)$ norm, is at most

$$E_{n,N,h} = C \left\{ e^{-\omega N} + h^s + \sqrt{\frac{L_n}{n h^{2m+2}}} + \frac{L_n}{n h^{m+2}} + n^{-2} + \delta h^{-m-3} \right\} \quad (7.12)$$

with probability at least $1 - \zeta$. This lemma allows bounded adversarial readout errors after acceptance; it does not change event tags or replace the fixed offsets by varying offsets.

For clarity, include the nuisance parameter in the realizability step. The subset $\{(f_{b,d_b}^T|_Q)_b : (T, d) \in \mathcal{U}\}$ of the finite product $C^m(Q)^J$ is separable. Choose a countable dense subset of that image and one representative $(T_j, d^j) \in \mathcal{U}$ for each of its points. The infimum of the distances from the fitted tuple to these points is a measurable function. Select the first point whose distance is within n^{-2} of that infimum. Such a point exists, the selection is measurable, and its distance from the true law tuple is at most $2E_{n,N,h} + n^{-2}$. Apply (7.10) only to this selected realizable pair. Require $2E_{n,N,h} + n^{-2} \leq \eta_0$; this is the small-error condition. The output includes the representative's actual offset tuple, not an unconstrained clock estimate attached to an unrelated table.

Take

$$h = \max\{(L_n/n)^{1/(2(m+s)+2)}, \delta^{1/(m+s+3)}, n^{-1}\} \leq h_0.$$

This is the upper bandwidth condition. Substitution into (7.12), with constants enlarged in the small-error regime, gives $C\{e^{-\omega N} + (L_n/n)^\alpha + \delta^\gamma\}$. The first n_B successful marks are independent with their conditional densities: independent reset records admit the Bernoulli/conditional-mark representation used in Section A. The binomial lower-tail bound and a union bound show that all groups have n_B successes except on an event of probability at most ζ . If a group fails this requirement, choose one fixed representative of $\mathcal{U}$; otherwise apply the preceding selection to the first n_B successes. This defines an estimator for every record.

Finally the specified returning length satisfies

$$e^{-\omega N_B} \leq C(L_B/B)^\beta, \quad n_B \geq cB^{\omega/(\omega+\alpha\Gamma)} L_B^{\alpha\Gamma/(\omega+\alpha\Gamma)}.$$

The floor changes constants only, under the stipulated lower-count condition. Since $n_B \leq B$ and $L_{n_B} \leq CL_B$, one has $(L_{n_B}/n_B)^\alpha \leq C(L_B/B)^\beta$. Adding the two exceptional-event probabilities proves (7.11). At fixed confidence all count conditions hold for large B ; the bandwidth and small-error conditions still require sufficiently small readout error. They are not suppressed for varying confidence. $\square$

The two mathematical interfaces of the original theorem remain essential. Without a differentiated *relative* physical limit, (7.4) could not be estimated from rare finite bridges. Without the actual envelope inverse, recovery of the action functions would not identify flat changes of a smooth contact. The new algebra removes a second offset and its supplied numerical calibration; it is not presented as a replacement for those analytic arguments. The marks, exact tags and gates, finite-smoothness priors and independent resetting remain part of the observation problem.

8. THE COMPLETE INVARIANT OF THE CHARGED LOCAL EXPERIMENT

The conditional laws determine the reflecting germs and the offsets, but conditioning removes the normalization of phase volume. The charged experiment retains it. We identify the exact

information added by the success probabilities and prove a finite-preparation estimate for the resulting invariant. In particular the reconstructed area will belong to the same actual table as the reconstructed contacts and offsets.

Throughout this section the marked lattice and polygon are fixed, all physical contacts are distinct, and $p_b \neq 0$. Offsets $d_b > 0$ are fixed nuisance parameters as in Section 7. Put $\vartheta = (T, d)$ and write $\mathcal{A}(T) > 0$ for the free area of its marked cell. A failure records only $\dagger$. Every preparation is an independent draw from normalized Liouville measure. These conventions are unchanged when a success probability is used as inverse data.

8.1. An exact anchor for phase volume. For a returning $N = nP$, denote the success probability at phase b by $\pi_{N,b}^\vartheta$ and its conditional endpoint density by $f_{N,b}^\vartheta$. The density is normalized on the actual success gate Q_+ , not on a later interior restriction. The superscript will be suppressed when there is no ambiguity. The reference twist $D_{N,b}$ is computed in the returning signed coordinates of Lemma 3.1.

Proposition 8.1 (The phase-volume anchor). *For every permitted positive-residual gate containing the origin,*

$$\pi_{N,b} f_{N,b}(0,0) = \frac{D_{N,b} d_b}{2\pi \mathcal{A}(T)}, \quad \mathcal{A}(T) = \frac{D_{N,b} d_b}{2\pi \pi_{N,b} f_{N,b}(0,0)}. \quad (8.1)$$

The identities hold at finite N , without a limiting approximation. Once the limiting conditional laws have determined the contact germs and offsets, the right side contains no supplied curvature or amplitude. It uses the success probability and central conditional density of one finite experiment.

Alternatively, at any one fixed phase the additional scalar

$$\ell_b := \lim_{n \rightarrow \infty} \frac{\pi_{nP,b}}{D_{nP,b}} = \frac{d_b}{2\pi \mathcal{A}(T) f_b(0,0)} \quad (8.2)$$

exists and determines the area together with the limiting laws. Here $D_{nP,b}$ is reconstructed from the contact curvatures, not supplied as additional data.

Proof. The success subdensity in (6.3) is

$$\frac{D_{N,b}}{2\pi \mathcal{A}(T)} \beta_{N,b}(u,v) r_{N,b,d_b}(u,v).$$

At the origin $\beta_{N,b}(0,0) = 1$ exactly, by the definition of the reference twist. The gauged action is zero there, so $r_{N,b,d_b}(0,0) = d_b$. The subdensity is also $\pi_{N,b} f_{N,b}$, proving (8.1). The relative law gives $f_{nP,b}(0,0) \rightarrow f_b(0,0) > 0$; division in the first identity proves (8.2). Theorem 7.3 determines d_b and the actual contact germs. Their curvatures and the marked flight lengths determine the Jacobi matrices and hence $D_{N,b}$ by (2.4). No unknown normalization survives. $\square$

There is an equivalent raw exponential normalization. With χ and $\mathcal{M}_b$ reconstructed from those same curvatures,

$$\lim_{n \rightarrow \infty} e^{n\chi} \pi_{nP,b} = \frac{d_b \sinh \chi}{\pi \mathcal{A}(T) (\mathcal{M}_b)_{12} f_b(0,0)}. \quad (8.3)$$

This follows because $e^{n\chi} D_{nP,b} \rightarrow 2 \sinh \chi / (\mathcal{M}_b)_{12}$. The prefactor, not just the exponential decay exponent, is needed for area recovery. Neither (8.2) nor (8.3) is an observation of an infinite limit from a finite sample. A charged finite-sample construction is given below.

8.2. The fibers of the observation map. We use germs of the local experiment in the precise sense of Section 6. Two experiments agree as germs when they agree on all sufficiently small common positive gates and all sufficiently long returning lengths, with the gate normalization specified anew. Equivalently one may restrict to a cofinal family of such gates. This qualification does not assert that equality of contact germs identifies trajectories whose endpoints lie outside their common representatives. Write

$$\mathfrak{J}(\vartheta) = ((F_b^T)_b \text{ as germs}, (d_b)_b, \mathcal{A}(T)).$$

Theorem 8.2 (Complete local observation invariant). *For actual marked tables in the stated class, equality of the germs of all charged reset records is equivalent to equality of $\mathfrak{J}(\vartheta)$. Equality of the single-offset limiting conditional law germs alone is equivalent to equality of the first two components of this invariant.*

In addition to the limiting law germs, the scalar (8.2) at one phase and one fixed common gate suffices to determine the remaining component. The finite alternative (8.1) also suffices. No remote boundary profile is included in this conclusion.

Proof. Equality of the charged records gives equality of their success masses and, since these are positive, equality of their conditional densities. Taking the relative limit on each common gate and applying Theorem 7.3 identifies the offsets and all contact germs. In particular both candidates have the same quadratic Jacobi operator, and therefore the same $D_{N,b}$. Their common success subdensity at the origin in any one finite experiment identifies their areas by (8.1).

Conversely, suppose the three components of $\mathfrak{J}$ agree. Choose representatives of the common contact germs and shrink the gates so that all stationary bridges and allowed end segments stay in these collars and their clear trajectory tubes. The common margins and weighted uniqueness used in Proposition 6.1 apply. That proposition's proof uses only equality of the offset in each experiment, not that the offsets were reported to the observer. It therefore gives identical success submeasures at every phase and returning length. Equal areas remove its sole multiplicative factor. The failure masses then agree as well. This proves the charged-record assertion.

For limiting conditional laws, the forward implication is exactly Theorem 7.3. The converse follows from the same locality calculation: the area factor cancels upon conditioning, whether or not the areas agree. Finally Proposition 8.1 proves both assertions about one additional scalar record. $\square$

The area component is independent of the conditional one. A smooth bump on an unvisited boundary arc, disjoint from the fixed collars and trajectory tubes, can have nonzero first area variation. For sufficiently small parameter it preserves strict convexity, clearance and all marks, while changing free area. The conditional experiments remain identical, but their charged success masses differ by the exact area ratio. With the second compensating bump of Corollary 6.3, the area can instead be held exactly fixed. That nonconstant family has the same complete charged-record germ. Thus the theorem describes both the recovered global scalar and the precise surviving freedom in remote completion; it does not infer a whole smooth table from local data.

The equivalence also holds for every fixed finite adaptive composition whose settings are chosen from these local experiments by a common record-based rule and independent randomization. Equality of the one-step record kernels proves this by conditioning on the past. The deterministic one-step experiments are among these compositions, so the converse is unchanged. This statement is about equality of experiment families, not about a sufficient statistic for one fixed finite sample.

8.3. Sensitivity of the rare normalization. Area reconstruction requires relative control of the reference twist. An error in curvature cannot be propagated through N flights with a constant independent of N . The following estimate retains that cost. Write $c_i = F_i''(0)$, fix

$k_i = L_i^{-1}$, and use the sign-conjugated transfer matrices with masses $2c_i$. All derivatives below are with respect to the finite vector c .

Lemma 8.3 (Logarithmic twist comparison). *On a fixed compact positive curvature box there is C such that, for all its $c, \tilde{c}$, phases b , and $N = nP$,*

$$\left| \log \frac{D_{N,b}(\tilde{c})}{D_{N,b}(c)} \right| \leq C(1+N)\|\tilde{c} - c\|_\infty. \quad (8.4)$$

More precisely,

$$\nabla \log D_{N,b} = \{\coth \chi - n \coth(n\chi)\} \nabla \chi - \nabla \log(\mathcal{M}_b)_{12}. \quad (8.5)$$

Along a smooth curvature path, if $\dot{\chi} \neq 0$, then $n^{-1} \partial_t \log D_{nP,b} \rightarrow -\dot{\chi}$. Thus the flight-length factor cannot in general be deleted from this normalization estimate. This last assertion is not a minimax lower bound.

Proof. Every matrix

$$Q_i(c_i) = \begin{pmatrix} 1 + 2c_i/k_i & 1/k_i \\ 2c_i & 1 \end{pmatrix}$$

has determinant one and positive entries. On the compact positive box, $\chi(c) = \operatorname{arcosh}(\operatorname{tr} \mathcal{M}_b(c)/2)$ is smooth with a positive lower bound, and $(\mathcal{M}_b)_{12}$ has a positive lower bound. Their first derivatives are bounded. Taking the logarithm of (2.4) and differentiating proves (8.5). Since $\coth(n\chi) \leq \coth \chi_*$ for $n \geq 1$, the gradient is bounded by $C(1+n)$. Integrating over the straight segment between the curvature vectors proves (8.4). Dividing the derivative identity by n proves the final limit. There are admissible directions with $\dot{\chi} \neq 0$: increasing one positive curvature makes the trace of the positive monodromy increase strictly. This follows directly by differentiating the displayed positive matrix product. A sufficiently small local curvature bump realizes such a direction while preserving the marked polygon and its open margins. $\square$

We next turn the identity into an estimate on realizable finite records. Use the bounded class $\mathcal{U}$ of Section 7, with $m \geq 3$ chosen so that $6a^m/(1-a^m) < 1$ on the enlarged positive quadratic class, and the same finite forward derivative bounds. Choose $Q \subseteq Q_+$ within the common stability collar, with $0 \in \operatorname{int} Q$. Densities below are normalized on Q_+ and restricted to Q without renormalization. Fix one phase b_0 . The forward density floors are uniform for $N \geq N_0$, after increasing N_0 if necessary.

Proposition 8.4 (Stability on the finite-record image). *For $\vartheta = (T, d)$ and $\tilde{\vartheta} = (\tilde{T}, \tilde{d})$ in $\mathcal{U}$, put*

$$\Delta_f = \max_b \|f_{N,b}^\vartheta - \tilde{f}_{N,b}^{\tilde{\vartheta}}\|_{C^m(Q)},$$

$$\Delta_\pi = \left| \log \frac{\pi_{N,b_0}^{\tilde{\vartheta}}}{\pi_{N,b_0}^\vartheta} \right|, \quad e_N = \Delta_f + e^{-\omega N}.$$

There are class constants $C, \eta_1 > 0$ such that $e_N \leq \eta_1$ implies

$$\|F^T - F^{\tilde{T}}\|_{C^0(I)} + \|d - \tilde{d}\|_\infty + \|c - \tilde{c}\|_\infty \leq C e_N, \quad (8.6)$$

$$\left| \log \frac{\mathcal{A}(\tilde{T})}{\mathcal{A}(T)} \right| \leq C(1+N)e_N + \Delta_\pi. \quad (8.7)$$

An upper bound for the areas also gives the corresponding absolute area estimate. No absolute error is divided by an exponentially small twist.

Proof. The uniform relative comparison bounds the distance between the two limiting law tuples by $\Delta_f + 2C_0e^{-\omega N}$. Make η_1 small enough that Theorem 7.3 applies. Its proof first applies the fixed-slice separation to obtain C^m control of the future actions. Their second derivatives, followed by the global positive quadratic estimate (4.6), control $c - \tilde{c}$ as well. This proves (8.6).

Apply the finite identity (8.1) at b_0 to both actual candidates. Subtraction of its logarithms gives

$$\begin{aligned} \log \frac{\mathcal{A}(\tilde{T})}{\mathcal{A}(T)} &= \log \frac{D_{N,b_0}(\tilde{c})}{D_{N,b_0}(c)} + \log \frac{\tilde{d}_{b_0}}{d_{b_0}} - \log \frac{\pi_{N,b_0}^{\tilde{\vartheta}}}{\pi_{N,b_0}^{\vartheta}} \\ &\quad - \log \frac{f_{N,b_0}^{\tilde{\vartheta}}(0,0)}{f_{N,b_0}^{\vartheta}(0,0)}. \end{aligned}$$

The positive offset and density floors control the second and fourth terms by Ce_N . Lemma 8.3 and (8.6) control the first term. This proves (8.7). If both areas are at most $\mathcal{A}_+$, the mean value theorem for the exponential gives $|\mathcal{A}(\tilde{T}) - \mathcal{A}(T)| \leq \mathcal{A}_+ |\log \mathcal{A}(\tilde{T}) - \log \mathcal{A}(T)|$. $\square$

8.4. A jointly realizable estimator. The count at phase b_0 is already part of the charged record; it does not require another preparation group. To keep the area and contacts compatible, use the finite-record image

$$\mathcal{J}_N(\vartheta) = ((f_{N,b}^{\vartheta}|_Q)_b, \log \pi_{N,b_0}^{\vartheta}) \in C^m(Q)^J \times \mathbb{R}, \quad J = r, \quad (8.8)$$

with the sum of the maximum density norm and the scalar absolute value. For each returning N choose a countable dense subset of this image and retain one actual representative $\vartheta_j \in \mathcal{U}$ at each point. Such a subset exists because the ambient metric space is separable. The representatives are not required to exhaust the tables.

For B resets in each phase, let S_b be its success count and put $q_N = c_0e^{-\Gamma N}$ and $n = \lfloor Bq_N/2 \rfloor$. On the event $S_b \geq n$ for every phase, form the interior kernel estimates $\hat{f}_{N,b}$ from the first n successful marks and put $\hat{\pi} = S_{b_0}/B$. Select the first representative whose distance from $((\hat{f}_{N,b})_b, \log \hat{\pi})$ is within n^{-2} of the infimum over the countable image. On the complementary event choose one fixed member of $\mathcal{U}$. This defines a measurable estimator $(\hat{T}, \hat{d})$ for every record, with its area $\mathcal{A}(\hat{T})$ taken from that same representative. It is not an efficient enumeration algorithm or a projection onto an arbitrary factor-model image.

Theorem 8.5 (Charged recovery of the complete invariant). *In the stated bounded class, set*

$$\alpha = \frac{s}{2(m+s)+2}, \quad \gamma = \frac{s}{m+s+3}, \quad L_n = \log(C_1 J n / \zeta), \quad 0 < \zeta < \frac{1}{2}.$$

Suppose $N \geq N_0$, $n \geq 2$, $Bq_N \geq C_2 \log(4J/\zeta)$ and $L_n/n \leq 1$. Use

$$h = \max\{(L_n/n)^{1/[2(m+s)+2]}, \delta^{1/(m+s+3)}, n^{-1}\} \leq h_0.$$

For $e^{-\omega N} + (L_n/n)^\alpha + \delta^\gamma$ sufficiently small, the estimator above satisfies, with probability at least $1 - 2\zeta$,

$$\begin{aligned} &\|F^{\hat{T}} - F^T\|_{C^0(I)} + \|\hat{d} - d\|_\infty + \left| \log \frac{\mathcal{A}(\hat{T})}{\mathcal{A}(T)} \right| \\ &\leq C(1+N)\{e^{-\omega N} + (L_n/n)^\alpha + \delta^\gamma\}. \end{aligned} \quad (8.9)$$

All $JB = rB$ resets, including failures, are charged. Readout errors are bounded by δ after acceptance; the tags and gates remain exact. Only the count at b_0 is used as additional geometric information. The estimate is a sufficient upper bound, not an optimality assertion.

Proof. For a binomial count with success probability $\pi_b \geq q_N$, Bernstein's inequality gives, for $t > 0$, a failure probability at most $2e^{-t}$ for

$$\left| \frac{S_b/B}{\pi_b} - 1 \right| \leq \sqrt{\frac{2t}{B\pi_b}} + \frac{2t}{3B\pi_b}.$$

Take $t = \log(4J/\zeta)$. Increasing the absolute constant C_2 makes the right side at most $1/2$ simultaneously for all phases, except on an event of probability at most ζ . On that event $S_b \geq B\pi_b/2 \geq n$, and $|\log(1+x)| \leq 2|x|$ gives

$$|\log \hat{\pi} - \log \pi_{N,b_0}| \leq C\sqrt{t/(Bq_N)}. \quad (8.10)$$

Independent resetting admits the Bernoulli/conditional-mark representation from Section A. Thus the first n successful latent marks have the required conditional law. The density estimate can be justified on their infinite extension even when the finite record has fewer successes; the count event ensures that only actually observed marks are used. No independence between the count error and the density error is required for the union bound.

Lemma A.1, before adding any physical limit bias, gives a simultaneous finite-density error at most

$$C \left\{ h^s + \sqrt{\frac{L_n}{nh^{2m+2}}} + \frac{L_n}{nh^{m+2}} + n^{-2} + \delta h^{-m-3} \right\}$$

except on an event of probability at most ζ . The displayed bandwidth bounds this by $C\{(L_n/n)^\alpha + \delta^\gamma\}$. Because n is comparable to Bq_N under $n \geq 2$, and $t \leq CL_n$, (8.10) has this same upper bound: $\alpha < 1/2$ and $L_n/n \leq 1$. The fitted finite image is consequently within $C\{(L_n/n)^\alpha + \delta^\gamma\}$ of the true image.

Density of the countable image and the near-minimum selection show that the distance between the selected and true images is at most twice this error plus n^{-2} . Both images correspond to actual pairs in $\mathcal{U}$, including their actual areas. Apply Proposition 8.4, adding only now the forward term $e^{-\omega N}$. The stipulated small-error condition ensures its applicability. The two exceptional events have total probability at most 2ζ , proving the theorem. There is no independent numerical area estimate attached to an unrelated completion. $\square$

Corollary 8.6 (A budget and readout dependent returning length). *Let*

$$\beta = \frac{\alpha\omega}{\omega + \alpha\Gamma}, \quad L_B = \log(C_*JB/\zeta), \quad t_{B,\delta} = (L_B/B)^\beta + \delta^\gamma < 1,$$

and choose

$$N = P \left\lfloor \frac{\log(1/t_{B,\delta})}{P\omega} \right\rfloor, \quad n = \lfloor Bc_0e^{-\Gamma N}/2 \rfloor. \quad (8.11)$$

Whenever the count, bandwidth, returning-length and small-error conditions of Theorem 8.5 hold, its right side is at most

$$Ct_{B,\delta} \log(e/t_{B,\delta}). \quad (8.12)$$

At zero readout error the returning-length choice is the same as in Theorem 7.5. At positive readout error it stops the length before curvature error is unnecessarily amplified. The original profile-and-offset rate (7.11) is retained separately; the logarithmic factor here pays for the additionally reconstructed area.

Proof. Write $x = (L_B/B)^\beta$ and $t = t_{B,\delta} \geq x$. The floor gives $e^{-\omega N} \leq e^{\omega P}t$ and $1+N \leq C \log(e/t)$. Also $e^{-\Gamma N} \geq t^{\Gamma/\omega}$, so the lower-count condition gives $n \geq cBt^{\Gamma/\omega}$. Since $n \leq B$, $L_n \leq CL_B$, and hence

$$(L_n/n)^\alpha \leq C(L_B/B)^\alpha t^{-\alpha\Gamma/\omega} \leq C(L_B/B)^\alpha x^{-\alpha\Gamma/\omega} = Cx \leq Ct.$$

The equality is the defining identity for β . Finally $\delta^\gamma \leq t$. Substitution into (8.9) proves (8.12). When $\delta = 0$, $(\log(1/x))/\omega = \alpha \log(B/L_B)/(\omega + \alpha\Gamma)$, giving the last comparison. $\square$

The complete invariant is therefore accessible without numerical clock calibration, supplied curvature or separately calibrated local flux amplitudes. The area assertion uses normalized unit-speed Liouville preparations and exact acceptance tags. Its contact part uses the actual smooth inverse, its volume part uses the unconditioned phase normalization, and their finite reconstruction uses relative rather than absolute rare-event control. The classification and estimate concern this specified marked reset experiment. They neither identify it with an exterior scattering observation nor turn its nontrivial equal-area completion fibers into determined boundary arcs.

9. RECONSTRUCTION FROM FIRST-MOMENT PROFILES

The separated limiting law has an additional consequence for the amount of smoothing required by the inverse problem. Four functions of one variable determine the two-variable law. The relevant moment matrix is nonsingular for a geometric reason: the two marked first derivatives have opposite signs. We first prove this assertion on the limiting model, then retain the finite-flight error, and finally construct an estimator on the image of actual tables. All samples in this section are the same paired endpoint records as before.

9.1. An exact moment identity. Fix a symmetric interval $J = [-R_+, R_+]$ and an interior interval $I = [-R, R] \Subset J$, both containing zero. For an integrable kernel f on J^2 define

$$M_j(u) = \int_J v^j f(u, v) dv, \quad N_j(v) = \int_J u^j f(u, v) du, \quad j = 0, 1, \quad (9.1)$$

$$H_f = \int_{J^2} \begin{pmatrix} 1 \\ u \end{pmatrix} (1 \ v) f(u, v) du dv. \quad (9.2)$$

Write $M(u) = (M_0(u), M_1(u))$ and $N(v) = (N_0(v), N_1(v))^\top$. The compression used below is

$$\mathcal{C}_I(f) = (M_0|_I, M_1|_I, N_0|_I, N_1|_I, H_f) \in \mathcal{X}_m := C^m(I)^4 \times \mathbb{R}^{2 \times 2}.$$

Equip this space with the sum of the largest function norm and the matrix operator norm. The integration domain in all five components is the full gate J^2 . In particular, $H_{f,00} = 1$ for a density normalized on that gate; restriction to I does not renormalize it. The functions M_1, N_1 retain paired information and are not ordinary marginal densities.

Lemma 9.1 (Moment reconstruction). *Suppose $f(u, v) = a(u)^\top C b(v)$, where a, b have two components and C is a 2×2 matrix. If H_f is nonsingular, then*

$$f(u, v) = M(u) H_f^{-1} N(v), \quad (u, v) \in J^2. \quad (9.3)$$

For each integer $m \geq 0$, the map

$$\mathcal{R} : (M_0, M_1, N_0, N_1, H) \longmapsto [(u, v) \mapsto M(u) H^{-1} N(v)]$$

is uniformly Lipschitz into $C^m(I^2)$ on sets with bounded $\mathcal{X}_m$ norm and $|\det H| \geq \kappa_* > 0$. Here $C^m(I^2)$ uses derivatives of total order at most m .

Proof. Put

$$L = \int_J \begin{pmatrix} 1 \\ u \end{pmatrix} a(u)^\top du, \quad R_0 = \int_J b(v) (1 \ v) dv.$$

Fubini's theorem gives $H_f = L C R_0$, $M(u) = a(u)^\top C R_0$, and $N(v) = L C b(v)$. Nonsingularity of their square product implies that all three factors are nonsingular. Cancelling them proves (9.3). This argument also proves that the separation rank is exactly two under the hypothesis.

On the stated bounded set, the adjugate formula bounds $\|H^{-1}\|$. For a second matrix $\tilde{H}$ use

$$H^{-1} - \tilde{H}^{-1} = H^{-1}(\tilde{H} - H)\tilde{H}^{-1}.$$

Add and subtract the three products obtained by replacing one factor at a time. For $i + j \leq m$, differentiation of a product gives $M^{(i)}(u)H^{-1}N^{(j)}(v)$. The same bounds therefore control every required derivative. No derivative of the data above order m is used. $\square$

This is a continuous moment version of the classical exact low-rank matrix reconstruction underlying CUR decompositions; see [7]. Its use here requires, in addition, a nondegeneracy bound forced by the billiard law and control of the defect of finite-flight laws from the separated model.

Proposition 9.2 (Geometric nondegeneracy). *Let the limiting law at one phase be*

$$f(u, v) = Z^{-1}w(u)z(v)\{d - A(u) - C(v)\}, \quad (u, v) \in J^2, \quad (9.4)$$

with the hypotheses of (7.1)–(7.2). Suppose throughout J that A' has sign $-p$, C' has sign p , and $|A'|, |C'| \geq p_/2 > 0$. Then $\det H_f > 0$ and (9.3) holds.*

On a bounded class with $|p| \geq p_$, fixed J , uniform positive lower and upper bounds for w, z , and an upper bound for Z , the determinant has a positive uniform lower bound. The derivative hypotheses follow by shrinking J once from the uniform C^2 action bounds and $A'(0) = -p$, $C'(0) = p$. They are not additional measured amplitudes or supplied curvature data.*

Proof. Use

$$a(u) = \begin{pmatrix} w(u) \\ w(u)A(u) \end{pmatrix}, \quad b(v) = \begin{pmatrix} z(v) \\ z(v)C(v) \end{pmatrix}, \quad C_0 = Z^{-1} \begin{pmatrix} d & -1 \\ -1 & 0 \end{pmatrix}$$

in Lemma 9.1. To compute the determinant without assuming its nonvanishing, set $W = \int_J w$, $V = \int_J z$, and let μ_w, μ_z have densities $w/W, z/V$. Direct evaluation gives

$$\det H_f = -\frac{W^2 V^2}{Z^2} \text{Cov}_{\mu_w}(u, A(u)) \text{Cov}_{\mu_z}(v, C(v)). \quad (9.5)$$

Indeed the two covariance factors, multiplied by W^2, V^2 , are respectively $\det L, \det R_0$, and $\det C_0 = -Z^{-2}$. For a probability measure μ and independent X, Y of law μ ,

$$\text{Cov}_{\mu}(x, g(x)) = \frac{1}{2} \mathbb{E}[(X - Y)(g(X) - g(Y))].$$

Consequently a derivative of constant sign and magnitude at least $p_*/2$ gives a covariance of that sign and magnitude at least $(p_*/2) \text{Var}_{\mu}(x)$. The two covariances in (9.5) have opposite signs.

For completeness, if $w_* \leq w \leq w^*$ on J , then

$$\text{Var}_{\mu_w}(u) \geq \frac{w_*}{2R_+ w^*} \inf_{c \in \mathbb{R}} \int_{-R_+}^{R_+} (u - c)^2 du = \frac{w_*}{w^*} \frac{R_+^2}{3}.$$

The same estimate applies to z , while W, V have positive lower bounds. Substitution in (9.5) proves the uniform claim. Finally, if the second action derivatives are bounded by K_2 , choose $K_2 R_+ \leq p_*/2$. The mean value theorem gives the asserted sign and magnitude bounds. All constants may deteriorate as the fixed gate shrinks or p_* tends to zero; no uniformity in those limits is claimed. $\square$

Remark 9.3 (Explicit fixed-gate conditioning). On $J = [-R, R]$, let $p \neq 0$ and $d > 2|p|R$. The positive normalized rank-two model

$$f(u, v) = \frac{d + pu - pv}{4R^2 d}$$

has moments

$$\mathbb{E}U = \frac{pR^2}{3d}, \quad \mathbb{E}V = -\frac{pR^2}{3d}, \quad \mathbb{E}(UV) = 0.$$

Indeed the odd integrals vanish and $\int_{-R}^R u^2 du = 2R^3/3$. Hence

$$H_f = \begin{pmatrix} 1 & -pR^2/(3d) \\ pR^2/(3d) & 0 \end{pmatrix}, \quad \det H_f = \frac{p^2 R^4}{9d^2}. \quad (9.6)$$

Thus even this simple model loses the determinant at fourth order in the gate radius and second order in obliquity. For $a = pR^2/(3d)$, $H_f^{-1} = \begin{pmatrix} 0 & 1/a \\ -1/a & 1/a^2 \end{pmatrix}$, so the inverse-matrix norm also diverges. This is an algebraic model, not an assertion that affine half-line actions are realized by a positive-curvature billiard. It illustrates why the fixed-class constants in Proposition 9.2 retain their gate and slope dependence. Rescaling coordinates changes moment units; it does not prove uniform physical stability in a degenerating class. The affine computation was supplied in the author-requested memorandum [8]; it is included here with its derivation.

Corollary 9.4 (Moment determination of the local invariant). *For the oblique marked class, four limiting moment profiles and their matrix H_f at each phase determine the offsets and all smooth contact germs. Adding the charged success probabilities as a function of the returning length determines the free area. Conversely, common contact germs, offsets and free area give identical compressed charged local experiments on sufficiently small common gates.*

Proof. Choose the common gates as in Proposition 9.2. The moment identity reconstructs each limiting law on I^2 . Theorem 7.3 determines the offsets and smooth contact germs. Their curvatures give $D_{N,b}$, and (8.2) then determines the area from one phase. The converse is the locality direction of Theorem 8.2, followed by integration. Applying this argument on a cofinal collection of gates gives the assertion for germs. $\square$

At finite N the identity in Lemma 9.1 need not hold for f_N . Thus the corollary concerns a family of laws and its relative limit, not a sufficient statistic for one finite sample. Nor does it replace paired endpoints by two unpaired marginal samples. For example, at normal incidence the positive model $Z^{-1}(d - au^2 - cv^2)$ on a symmetric square can have rank two but $M_1 = N_1 = 0$, so H_f is singular. The two-offset results at normal incidence remain available independently of the first-moment inverse.

9.2. Finite-flight comparison on the physical image. Fix the bounded class $\mathcal{U}$ of Proposition 8.4. In particular, $m \geq 3$ satisfies the explicit later-visit condition on the enlarged positive quadratic class. Choose $J, I \subseteq J$, and a contact collar $I_c \subseteq I$ once so that the preceding proposition and the inherited action-to-contact stability apply uniformly. Densities are normalized on the success gate J^2 ; all integrations below still use that whole gate. The relative physical theorem supplies

$$\max_b \|f_{N,b}^\vartheta - f_b^\vartheta\|_{C^m(J^2)} \leq C_0 e^{-\omega N}, \quad N \geq N_0, \quad (9.7)$$

uniformly over actual pairs $\vartheta = (T, d) \in \mathcal{U}$. Shrinking the fixed gates changes class constants, not the charged preparation convention.

Proposition 9.5 (Stability of compressed finite records). *For two actual pairs in $\mathcal{U}$ put*

$$\Delta_M = \max_b \|\mathcal{C}_I(f_{N,b}^\vartheta) - \mathcal{C}_I(\tilde{f}_{N,b}^\vartheta)\|_{\mathcal{X}_m}, \quad e_N = \Delta_M + e^{-\omega N}, \quad \Delta_\pi = \left| \log \frac{\pi_{N,b_0}^{\tilde{\vartheta}}}{\pi_{N,b_0}^\vartheta} \right|.$$

There are class constants $C, \eta_* > 0$ such that $e_N \leq \eta_*$ implies

$$\|F^T - \tilde{F}^T\|_{C^0(I_c)} + \|d - \tilde{d}\|_\infty + \|c - \tilde{c}\|_\infty \leq Ce_N, \quad (9.8)$$

$$\left| \log \frac{\mathcal{A}(\tilde{T})}{\mathcal{A}(T)} \right| \leq C(1 + N)e_N + \Delta_\pi. \quad (9.9)$$

Here c is the vector of contact curvatures. The estimate does not assume exact rank two at finite flight length.

Proof. Integration in (9.1)–(9.2) is a bounded linear map from $C^m(J^2)$ to $\mathcal{X}_m$. Hence the two limiting compressed tuples are at distance at most $\Delta_M + Ce^{-\omega N}$. Proposition 9.2 gives their uniform determinant floor. Lemma 9.1 therefore bounds the distance of the two limiting densities on I^2 by Ce_N . Equation (9.7) also gives

$$\max_b \|f_{N,b}^\vartheta - \tilde{f}_{N,b}^\vartheta\|_{C^m(I^2)} \leq Ce_N. \quad (9.10)$$

Now apply Proposition 8.4 on these fixed interior and contact collars. Its proof uses the C^m law norm to recover the actions and their quadratic coefficients before estimating the profiles. In particular it does not infer curvature control from C^0 profile control. The exact same-gate central-density identity and the logarithmic twist comparison then give (9.9).

Equivalently, the preceding argument shows that $\mathcal{R}(\mathcal{C}_I(f_{N,b}))$ approximates $f_{N,b}|_{I^2}$ with error $Ce^{-\omega N}$, for N large enough. This is only an approximation: the finite central density in the area identity is the one belonging to the actual candidate, not an exactly identified rank-two surrogate. $\square$

9.3. Estimation of the four profiles. We give the sampling estimate in detail because smoothing a weighted marginal, rather than a two-variable density, changes the exponents. Let $s \geq 1$ be an integer. Assume the same uniform forward bounds as before, now stated on the full fixed gate:

$$\sup_{N \geq N_0, b, \vartheta \in \mathcal{U}} \|f_{N,b}^\vartheta\|_{C^{m+s}(J^2)} \leq K.$$

All constants below may depend on this class, the fixed intervals, m, s and the number r of phases.

Choose $K_0 \in C_c^{m+1}((-1, 1))$ with integral one and vanishing moments of orders $1, \dots, s-1$. Such a kernel can be constructed by multiplying a positive smooth bump by a polynomial of degree at most $s-1$: the moment conditions are a nonsingular Gram system for polynomials on the support of that bump. The kernel need not be nonnegative. Let $h_0 \leq 1$ be smaller than the distance from I to ∂J .

For n accepted pairs at one phase, write their true coordinates as (U_i, V_i) and their readouts as $(\tilde{U}_i, \tilde{V}_i)$, with both coordinate errors at most $0 \leq \delta \leq 1$. Define

$$\widehat{M}_j(u) = \frac{1}{nh} \sum_{i=1}^n \tilde{V}_i^j K_0((u - \tilde{U}_i)/h), \quad (9.11)$$

$$\widehat{N}_j(v) = \frac{1}{nh} \sum_{i=1}^n \tilde{U}_i^j K_0((v - \tilde{V}_i)/h), \quad j = 0, 1, \quad (9.12)$$

$$\widehat{H} = \frac{1}{n} \sum_{i=1}^n \begin{pmatrix} 1 \\ \tilde{U}_i \end{pmatrix} (1 \quad \tilde{V}_i). \quad (9.13)$$

Denote the resulting tuple by $\widehat{\mathcal{C}}_{N,b}$. There is no inversion of a random nearly singular matrix in this definition.

Lemma 9.6 (One-dimensional smoothing bound). *For $n \geq 2$, $n^{-1} \leq h \leq h_0$, and $0 < \zeta < 1/2$, put $L_n = \log(C_1 r n / \zeta)$. The first n accepted latent pairs at each phase may be used in (9.11)–(9.13). With probability at least $1 - \zeta$,*

$$\max_b \|\widehat{\mathcal{C}}_{N,b} - \mathcal{C}_I(f_{N,b})\|_{\mathcal{X}_m} \leq C \left\{ h^s + \sqrt{\frac{L_n}{nh^{2m+1}}} + \frac{L_n}{nh^{m+1}} + n^{-2} + \delta h^{-(m+2)} \right\}. \quad (9.14)$$

The perturbations may depend arbitrarily on the latent sample. The probability statement is about the independent latent preparations; acceptance and its success/failure tags are exact.

Proof. First set $\delta = 0$. For $0 \leq k \leq m$, the expectation of $\widehat{M}_j^{(k)}(u)$ is the convolution of $M_j^{(k)}$ with the rescaled kernel on the interior interval. Integration by parts is legitimate because the kernel support does not meet ∂J . The C^{m+s} bound on f_N gives the same order bound on M_j . Taylor's formula and the moment cancellations give a uniform bias Ch^s . The argument for N_j is identical.

For a fixed u , a summand before division by n has magnitude at most Ch^{-k-1} and variance at most Ch^{-2k-1} . Indeed the integral of the square of $K_0^{(k)}((u - U)/h)$ against the bounded joint density gains one factor h ; $|V|^j$ is uniformly bounded on J . Bernstein's inequality thus gives, with failure probability at most $2e^{-t}$, fluctuation at most

$$C \left\{ \sqrt{\frac{t}{nh^{2k+1}}} + \frac{t}{nh^{k+1}} \right\}.$$

Take a grid in I of mesh at most $n^{-2}h^{m+2}$. Both the empirical derivative and its expectation have Lipschitz constants at most Ch^{-m-2} . Passage from the grid to the whole interval therefore costs at most Cn^{-2} . Since $h \geq n^{-1}$, the grid has at most Cn^{m+4} points. A union bound over the four profiles, r phases and $m+1$ derivative orders, with $t = C_m L_n$, proves the stated stochastic bound. The entries of $\widehat{H}$ are averages of bounded variables and have error at most $C\{\sqrt{L_n/n} + L_n/n\}$ on the same enlarged event. This is absorbed by the displayed bound when $h \leq 1$. Independence of the four profiles is neither assumed nor needed.

Restore the readout errors. The mean value theorem for the kernel and the bounded weights gives, for each summand of a derivative of order k ,

$$\left| \widetilde{V}_i^j h^{-k-1} K_0^{(k)}((u - \widetilde{U}_i)/h) - V_i^j h^{-k-1} K_0^{(k)}((u - U_i)/h) \right| \leq C\delta\{h^{-k-2} + h^{-k-1}\}.$$

The weights remain bounded by $C(R_+)$ because $\delta \leq 1$. For $k \leq m$ and $h \leq 1$ the last bound is $C\delta h^{-m-2}$. Averaging does not increase it. The matrix entries have deterministic perturbation at most $C\delta$, also absorbed. This comparison holds sample by sample and so requires no distributional assumption on the readout errors. $\square$

9.4. Charged recovery with one-dimensional smoothing. The actual finite image to be fitted is

$$\mathcal{K}_N(\vartheta) = ((\mathcal{C}_I(f_{N,b}^\vartheta))_{b=0}^{r-1}, \log \pi_{N,b_0}^\vartheta) \in \mathcal{X}_m^r \times \mathbb{R}. \quad (9.15)$$

Use the maximum of the tuple norms plus the scalar absolute value. For each returning N , fix a countable dense subset of this image, and one actual representative $(T_j, d_j) \in \mathcal{U}$ for each of its points. This is possible because $C^m(I)$, and hence the ambient metric space, is separable. Closedness of the image is not required.

Perform B independent normalized unit-speed Liouville resets at each phase. Put $q_N = c_0 e^{-\Gamma N}$ and $n = \lfloor Bq_N/2 \rfloor$. If every success count S_b is at least $n \geq 2$, use the first n accepted pairs for the estimated compressed tuples and set $\widehat{\pi} = S_{b_0}/B$. Select the first representative whose distance to $((\widehat{\mathcal{C}}_{N,b})_b, \log \widehat{\pi})$ is within n^{-2} of the infimum over the countable image. Otherwise select a fixed member of $\mathcal{U}$. The infimum of countably many measurable distances is measurable, and a

qualifying index exists by the definition of infimum. Thus this specifies an actual table-offset estimator on every record. Its area is always the area of that same table. An effective enumeration or polynomial-time search is not asserted.

Theorem 9.7 (Compressed charged reconstruction). *In the stated class set*

$$\alpha_1 = \frac{s}{2(m+s)+1}, \quad \gamma_1 = \frac{s}{m+s+2}. \quad (9.16)$$

Suppose $N \geq N_0$, $n \geq 2$, $Bq_N \geq C_2 \log(4r/\zeta)$, and $L_n/n \leq 1$. Choose

$$h = \max\{(L_n/n)^{1/[2(m+s)+1]}, \delta^{1/(m+s+2)}, n^{-1}\} \leq h_0.$$

For $\varepsilon_N = e^{-\omega N} + (L_n/n)^{\alpha_1} + \delta^{\gamma_1}$ sufficiently small, the estimator just defined satisfies, with probability at least $1 - 2\zeta$,

$$\|F^{\hat{T}} - F^T\|_{C^0(I_c)} + \|\hat{d} - d\|_\infty + \|\hat{c} - c\|_\infty \leq C\varepsilon_N, \quad (9.17)$$

$$\left| \log \frac{\mathcal{A}(\hat{T})}{\mathcal{A}(T)} \right| \leq C(1+N)\varepsilon_N. \quad (9.18)$$

All rB preparations, including failures, are charged. No additional phase group is needed for the area. Readout errors occur after exact acceptance and do not corrupt the success tags.

Proof. The count argument in (8.10), with $t = \log(4r/\zeta)$, gives simultaneously

$$S_b \geq n \quad \text{for every } b, \quad |\log \hat{\pi} - \log \pi_{N,b_0}| \leq C\sqrt{t/(Bq_N)}$$

except on an event of probability at most ζ . The Bernoulli/conditional-mark representation of independent resets gives independent latent marks of density $f_{N,b}$ at successive successes. They may be extended beyond the finite record when proving a probability bound; on the count event only the first n actually observed marks are used. This construction does not condition the marks on a possibly mark-dependent noisy acceptance test: acceptance is exact.

Apply Lemma 9.6 to these latent sequences. At the prescribed bandwidth, its right side is at most $C\{(L_n/n)^{\alpha_1} + \delta^{\gamma_1}\}$. For the variance term this follows by substituting the first bandwidth lower bound; the envelope exponent becomes $(m+2s)/[2(m+s)+1] \geq \alpha_1$. The readout term is bounded using the second lower bound, and the bias is bounded using the maximum formula. The n^{-2} term is absorbed. Since $\alpha_1 < 1/2$, n is comparable to Bq_N , and $L_n/n \leq 1$, the count error is absorbed by the same expression.

Density of the countable actual image implies that its infimal distance to the estimated tuple is no larger than the distance of the true image. The selected image is therefore within twice the estimation error plus n^{-2} of the true image. Both candidates are actual pairs. Proposition 9.5 gives the assertions, with the physical error $e^{-\omega N}$ added only at this step. The union of the count and smoothing exceptional events has probability at most 2ζ . $\square$

Corollary 9.8 (Budget-dependent flight length). *Let*

$$\beta_1 = \frac{\alpha_1 \omega}{\omega + \alpha_1 \Gamma}, \quad L_B = \log(C_* r B / \zeta), \quad t = (L_B/B)^{\beta_1} + \delta^{\gamma_1} < 1,$$

and choose $N = P[\log(1/t)/(P\omega)]$. Whenever the hypotheses of Theorem 9.7 hold, its local error is at most Ct and its joint local-and-area error is at most

$$Ct \log(e/t). \quad (9.19)$$

Proof. The floor gives $e^{-\omega N} \leq e^{\omega P} t$, $1 + N \leq C \log(e/t)$ and $n \geq cBt^{\Gamma/\omega}$ under the count conditions. Write $x = (L_B/B)^{\beta_1} \leq t$. Since $L_n \leq CL_B$,

$$(L_n/n)^{\alpha_1} \leq C(L_B/B)^{\alpha_1} t^{-\alpha_1 \Gamma/\omega} \leq C(L_B/B)^{\alpha_1} x^{-\alpha_1 \Gamma/\omega} = Cx.$$

The last equality is the definition of β_1 . Also $\delta^{\gamma_1} \leq t$. Substitution in (9.17)–(9.18) proves the claims. $\square$

For the same m, s, ω, Γ , the exponents α_1 and γ_1 strictly improve the two-dimensional smoothing exponents $s/[2(m+s)+2]$ and $s/(m+s+3)$ in Theorem 8.5. The corresponding budget exponent improves as well, since $\alpha \mapsto \alpha\omega/(\omega + \alpha\Gamma)$ is strictly increasing. These are comparisons of proved upper bounds on the same oblique bounded class, not minimax assertions or comparisons with lens or marked-length observations. The improvement comes from estimating one-dimensional weighted profiles, not from dividing a small absolute error by a rare success probability. The full smooth inverse and its finite-order later-visit estimate remain essential to the passage from these profiles to the actual reflecting contacts.

10. CONDITIONING OF THE PHYSICAL-TO-SMOOTH INVERSE

The inverse has three different quantitative costs: separating the physical actions, inverting their actual smooth response, and acquiring enough successful trajectories. They depend on different geometric margins. We collect them here on the fixed classes used in the preceding results. None of the following sufficient bounds is a lower bound for all possible reconstruction procedures.

10.1. Contraction, differentiability and phase weights. For the scalar visit bound $a < 1$, the sufficient weighted threshold and its least integer order are exactly

$$\theta_m < 1 \iff a^m < \frac{1}{7}, \quad m_{\min} = \max \left\{ 3, \left\lceil \frac{\log 7}{-\log a} \right\rceil + 1 \right\}. \quad (10.1)$$

At a chosen order the weighted inverse prefactor is $2/(1 - \theta_m)$. Minimizing the order need not give a comfortable inverse margin. For any chosen $0 < \vartheta < 1$, the choice

$$m \geq \max \left\{ 3, \left\lceil \frac{\log((6 + \vartheta)/\vartheta)}{-\log a} \right\rceil \right\} \quad (10.2)$$

ensures $\theta_m \leq \vartheta$ and hence a prefactor at most $2/(1 - \vartheta)$. These formulas follow by solving $6a^m/(1 - a^m) \leq \vartheta$; the strict inequality in (10.1) explains the floor followed by +1.

For example, the scalar bounds give the following values. The last column uses $s = 1$ in the one-dimensional smoothing exponent; it is not an asserted optimal statistical rate.

a	$m_{\min}$	$\theta_{m_{\min}}$	$2/(1 - \theta_{m_{\min}})$	α_1
1/2	3	0.857143	14	1/9
9/10	19	0.937099	31.7961	1/41
99/100	194	0.995514	445.8342	1/391

Taking instead $\vartheta = 1/2$ gives orders 4, 25, 256, respectively, and weighted prefactor at most 4. It increases the differentiability required of the observed action and the graph prior. The jet-alignment and forward constants also depend on this order; bounding the weighted prefactor alone does not bound the entire inverse.

Corollaries 5.8 and 5.9 replace a in these sufficient tests by $\bar{a} = (\prod_i \hat{a}_i)^{1/r}$ when phase-specific bounds are available on the enlarged class. The unweighted prefactor then contains $\kappa(w)$. The product of the phasewise upper bounds must not be replaced by the supremum of the products along one orbit without a new uniform estimate. Likewise an exactly computed linear multiplier does not itself provide the nonlinear collar bounds: the latter follow by continuity on the stipulated class. The scalar criterion remains valid and is often simpler to use.

10.2. Observation margins. In the two-offset quotient, put $h = A(u) + C(v)$ and $\Delta = d_2 - d_1 > 0$. On the exact image,

$$d_2 - d_1 Q = \frac{d_2 \Delta}{d_2 - h}, \quad \left| \frac{\partial H}{\partial Q} \right| = \frac{d_1 (d_2 - h)^2}{d_2 \Delta}.$$

Thus offset separation and the bounds for the positive residual enter the action inverse explicitly. Positive density floors are also required to form Q ; its higher derivatives involve the same reciprocal functions. This route does not require a nonzero obliquity p and therefore retains the normal-incidence case.

For a single law, the inverse additionally uses $\kappa_f = p^2/d^2$ and the fixed-slice anchors u_a, v_a of (7.6). The map $\kappa \mapsto |p|/\sqrt{\kappa}$ has derivative of absolute value $|p|/(2\kappa^{3/2})$. The geometric bounds after Theorem 7.2 supply $\kappa_f \geq p_*^2/d_+^2$ and $|u_a|, |v_a| \geq p_* a/(2D_*)$. Their degeneration is independent of the order condition for the smooth envelope.

Moment reconstruction brings its own matrix condition. In the affine model of Remark 9.3, put $z = pR^2/(3d)$. Then

$$H = \begin{pmatrix} 1 & -z \\ z & 0 \end{pmatrix}, \quad \det H = z^2, \quad H^{-1} = \begin{pmatrix} 0 & 1/z \\ -1/z & 1/z^2 \end{pmatrix}.$$

The shrinking-gate and small-obliquity losses are visible in the inverse, not just its determinant. This is an algebraic density example, not an asserted positive-curvature billiard realization. Rescaling the moments changes their units and the corresponding error norms; it does not remove the physical conditioning from a comparison in fixed units.

These requirements are separate from the positive Jacobi mass, nongrazing and clearance bounds that construct the stationary trajectories. They are also separate from the uniform C^{m+3} graph prior used to align jets. On a fixed class with all these margins, their contributions are finite; uniformity is not asserted as any of them degenerates.

10.3. Preparation budget and the exact finite anchor. For the moment protocol the conditional-sample and bounded-readout exponents are

$$\alpha_1 = \frac{s}{2(m+s)+1}, \quad \gamma_1 = \frac{s}{m+s+2}.$$

They arise from the complete bound in Lemma 9.6, not just its square-root term. In particular the Bernstein envelope, interpolation-grid and post-acceptance perturbation terms remain part of the estimate. Suppose the uniform finite-flight defect is $Ce^{-\omega N}$ and the success probability is at least $ce^{-\Gamma N}$. For B independent preparations per phase, the two terms to balance have the form

$$e^{-\omega N}, \quad \left(\frac{L_B e^{\Gamma N}}{B} \right)^{\alpha_1}.$$

Solving their equality gives

$$N \sim \frac{\alpha_1}{\omega + \alpha_1 \Gamma} \log \frac{B}{L_B}, \quad \beta_1 = \frac{\alpha_1 \omega}{\omega + \alpha_1 \Gamma}. \quad (10.3)$$

Rounding to a returning number of flights changes N by a bounded amount. The count, bandwidth, minimum-flight and small-error conditions in Theorem 9.7 and Corollary 9.8 must still hold. Increasing m may improve the weighted inverse margin while decreasing α_1 ; rarity decreases it further to β_1 . A positive exponent is not a bound for the computational cost of finding the selected actual table.

Counts enter area recovery through the exact finite identity

$$\mathcal{A}(T) = \frac{D_{N,b}(c)d_b}{2\pi\pi_{N,b}f_{N,b}(0,0)}.$$

The count and the fitted actual finite density use the same full gate. Replacing the latter by its rank-two limit would discard a finite-flight term. The logarithmic derivative of $D_{N,b}$ costs $1 + N$, as shown in Section 8; at (10.3) it produces the retained logarithmic area loss. Failures are charged, gate and itinerary tags are exact, and readout perturbations occur after acceptance. The independently reset Liouville experiment, not an arbitrary unknown preparation measure, supplies the normalizer. These are conditions on the stated inverse problem, not additional conclusions of the conditioning calculation.

Acknowledgments. The Riccati comparison leading to the positive quadratic inverse and the affine moment-conditioning example were communicated in the author-requested AI-assisted memoranda [3, 8]. These attributions concern the stated mathematical inputs, not journal endorsements.

APPENDIX A. SMOOTH CONTACT RECOVERY FROM FINITE PREPARATIONS

The preceding stability theorem requires a differentiated density norm. A finite endpoint record does not supply this norm, and a discontinuous histogram cannot be substituted for it. We now construct the missing statistical comparison, for the same marked periodic polygon and the same two controlled offsets. The target is the complete real contact profile, not a truncated jet. The proof regularizes the endpoint data, compares realizable laws, and then applies the geometric inverse. All independent phase preparations, including failures, are counted.

A.1. A fixed interior observation experiment. Fix the marked polygon of Section 3, with r physical phases, a returning oriented period P , and offsets $0 < d_1 < d_2$. Write $J = 2r$ for the number of phase–offset groups. Choose concentric squares

$$Q = [-R, R]^2 \Subset Q_+ = [-R_+, R_+]^2$$

inside one common positive-residual collar. A success means precisely the selected itinerary event of Theorem 2.5, with its recorded endpoints in Q_+ . This gate is fixed in the calibrated tangent coordinates; it uses no unknown normal graph height. The success/failure tag, including gate membership, is recorded exactly. The bounded errors below affect endpoint readout after acceptance, not selection tags or the two offsets. Conditional densities below are normalized on Q_+ , and compared on Q . Restricting them to Q does not renormalize them. The unknown scalar normalizers still cancel in Proposition 3.2.

Let $\mathcal{C}$ be a nonempty class of actual marked tables whose contact tuples belong to the bounded positive class $\mathcal{K}_m$ of (5.7). The integer $m \geq 3$ is fixed so that (5.4) holds on the enlarged quadratic class. Fix also an integer $s \geq 1$. Impose the uniform geometric hypotheses of Theorems 2.3 and 2.5 through endpoint order $m + s$, including the finitely many higher graph bounds used for that order, and a positive upper bound for the free cell area. Positivity, clearance, incidence, flight-length bounds, and the separated offsets are fixed throughout. These are forward smoothness and geometric priors, not bounds assumed for an observed density error. No analytic prior is imposed. A compact smooth family with the stated positive margins is one admissible choice; the constants use only its finitely stipulated derivative bounds.

For $T \in \mathcal{C}$, denote its profiles by F^T , the limiting density tuple by $f^T = (f_{b,d}^T)$, the N -flight conditional density by $f_{N,b,d}^T$, and its success probability by $p_{N,b,d}^T$. Here and below N is a multiple of P . By shrinking both squares within the stability collar, we obtain constants $H, C_0, c_0, \omega, \Gamma > 0$

and N_0 such that

$$\sup_{T,b,d,N \geq N_0} (\|f_{N,b,d}^T\|_{C^{m+s}(Q_+)} + \|f_{b,d}^T\|_{C^{m+s}(Q_+)}) \leq H, \quad (\text{A.1})$$

$$\max_{b,d} \|f_{N,b,d}^T - f_{b,d}^T\|_{C^m(Q_+)} \leq C_0 e^{-\omega N}, \quad (\text{A.2})$$

$$p_{N,b,d}^T \geq p_*(N) := c_0 e^{-\Gamma N}. \quad (\text{A.3})$$

We take $c_0 \leq 1$. To see the uniformity explicitly, the numerator of the conditional law is the normalized twist times its physical residual. The relative estimates bound its indicated derivatives; a fixed positive square bounds its integral away from zero. Division therefore gives the first two estimates. For the last one the exact reference twist is

$$D_{N,b} = \frac{\sinh \chi}{(\mathcal{M}_b)_{12} \sinh(N\chi/P)}.$$

The positive quadratic box bounds its prefactor below, and bounds χ/P above by a finite Γ . The residual integral has a positive lower bound and the area has an upper bound. The exact phase-volume formula gives (A.3). This argument uses relative twist control before conditioning; division of an absolute action error by $D_{N,b}$ would not give these estimates.

The law comparison in Theorem 5.5 now reads

$$\|F^T - \tilde{F}^T\|_{C^0([-R,R])} \leq C \max_{b,d} \|f_{b,d}^T - \tilde{f}_{b,d}^T\|_{C^m(Q)} \quad (\text{A.4})$$

when the right-hand density distance is at most a fixed η_0 . It applies to these restrictions because its proof uses the anchored quotient and positive denominator floors, not a known normalizing constant. A smaller common R can be chosen once for the class.

A.2. Recovering the differentiated density norm. Use the maximum of all derivatives of total order at most m as the $C^m(Q)$ norm. Fix a smooth compactly supported kernel K on $(-1,1)^2$ with

$$\int K = 1, \quad \int z^\alpha K(z) dz = 0 \quad (1 \leq |\alpha| < s).$$

Such kernels need not be nonnegative. For completeness, take a positive smooth bump w on $(-1,1)$, form the positive definite moment matrix $M_{ij} = \int t^{i+j} w(t) dt$, $0 \leq i, j < s$, and solve $Mc = (1, 0, \dots, 0)^t$. Then $k(t) = w(t) \sum_{j=0}^{s-1} c_j t^j$ has the required one-dimensional moments, and $K(z) = k(z_1)k(z_2)$ has the displayed moments. Positive definiteness follows by integrating the square of a nonzero polynomial against w . We write $K_h(z) = h^{-2}K(z/h)$.

Lemma A.1 (Interior derivative estimation). *Let each of J groups contain $n \geq 2$ independent latent endpoints $Y_{g,l}$ with density q_g on Q_+ and $\|q_g\|_{C^{m+s}(Q_+)} \leq H$. Observed endpoints may have arbitrary bounded recording errors $|\tilde{Y}_{g,l} - Y_{g,l}|_\infty \leq \delta$. Independence of the recording errors is not required. Set*

$$\hat{q}_{g,h}(z) = \frac{1}{n} \sum_{l=1}^n K_h(z - \tilde{Y}_{g,l}).$$

For $n^{-1} \leq h \leq h_0 < \min(1, R_+ - R)$ and $0 < \zeta < 1/2$, put $L_n = \log(C_1 J n / \zeta)$, with a sufficiently large fixed C_1 . With probability at least $1 - \zeta$,

$$\max_g \|\hat{q}_{g,h} - q_g\|_{C^m(Q)} \leq C \left\{ h^s + \sqrt{\frac{L_n}{nh^{2m+2}}} + \frac{L_n}{nh^{m+2}} + n^{-2} + \frac{\delta}{h^{m+3}} \right\}. \quad (\text{A.5})$$

The constants depend on the fixed squares, kernel, orders and H , not on n, h, δ or the particular densities.

Proof. First set $\delta = 0$. On Q the kernel does not encounter the boundary of Q_+ . Integration by parts gives $D^\alpha \mathbb{E} \hat{q}_h = K_h * D^\alpha q$ for $|\alpha| \leq m$. Taylor's integral remainder through order $s - 1$, with the moment cancellations, gives a uniform bias at most CHh^s . Only derivatives through $m + s$ on Q_+ are used; the zero extension of the density is not asserted to be smooth at the boundary.

For a fixed z and $|\alpha| \leq m$, the summand $D^\alpha K_h(z - Y)$ is bounded in absolute value by Ch^{-m-2} and has variance at most CHh^{-2m-2} . These bounds follow directly by scaling, since

$$\int |D^\alpha K_h(z - y)|^2 q(y) dy \leq Hh^{-2|\alpha|-2} \|D^\alpha K\|_2^2.$$

For independent centered summands bounded by U with variance at most V , Bernstein's bound for their average is

$$\mathbb{P}\left(\left|n^{-1} \sum_{l=1}^n Z_l\right| > \sqrt{2Vt/n} + 2Ut/(3n)\right) \leq 2e^{-t}.$$

One may derive it by expanding the exponential and using $\mathbb{E}|Z_l|^j \leq VU^{j-2}$ for $j \geq 2$: $\mathbb{E}e^{\lambda Z_l} \leq \exp\{\lambda^2 V/[2(1 - |\lambda|U/3)]\}$ for $|\lambda|U < 3$. Multiplication of the moment bounds and exponential Markov inequality give the displayed tail estimate, with harmless enlargement of U for centering.

Cover Q by a grid of mesh at most $n^{-2}h^{m+3}$. Its size is at most Cn^4h^{-2m-6} . A union bound over this grid, the fixed set of multiindices and the J groups, with $t = CL_n$, gives the square-root and linear terms of (A.5); here $h \geq n^{-1}$ absorbs the grid logarithm into CL_n . Both the empirical field and its expectation have Lipschitz constant at most Ch^{-m-3} after any derivative of order at most m . The grid error is consequently at most Cn^{-2} . This proves the asserted uniform stochastic bound. This elementary fixed-bandwidth argument is sufficient here; sharper classical uniform kernel results are discussed in [4].

Finally the mean value theorem, applied to each smooth kernel summand, gives the deterministic comparison

$$\max_g \|\hat{q}_{g,h}^{\text{observed}} - \hat{q}_{g,h}^{\text{latent}}\|_{C^m(Q)} \leq C\delta h^{-m-3}.$$

Adding it proves the result, even for data-dependent recording errors. The estimates need not be densities or nonnegative functions. No nonlinear law identity has yet been applied to them. $\square$

A.3. Realizable estimation and complete profile recovery. Let $\mathcal{F} = \{f^T|_Q : T \in \mathcal{C}\}$ in the product $C^m(Q)^J$, with maximum norm. Choose a countable observation-dense subset $\{f^{T_j} : j \geq 1\}$ of this image, retaining one actual table for each element. Such a subset exists because $C^m(Q)^J$ is separable and metrizable; its subspaces are separable. Given an estimated tuple $\hat{f}$ and a tolerance $\xi > 0$, define

$$\hat{T} = T_{j_*}, \quad j_* = \min \left\{ j : \|f^{T_j} - \hat{f}\|_{C^m(Q)} \leq \inf_{l \geq 1} \|f^{T_l} - \hat{f}\|_{C^m(Q)} + \xi \right\}. \quad (\text{A.6})$$

The set is nonempty by the definition of an infimum. This is a measurable estimator: each norm is a supremum of continuous derivative evaluations over a fixed countable dense set, and the remaining operations are countable. This existence construction is not a claim of efficient enumeration of all physical tables. It supplies no unknown curvature or flux amplitude as observation data.

Theorem A.2 (Finite-sample recovery of smooth contacts). *Under the stated class assumptions, collect n successful endpoints at N flights in each phase-offset group, from independent preparations. Allow recording error at most δ , and use (A.6) with the kernel estimates and $\xi = n^{-1}$. Let*

$$\alpha = \frac{s}{2(m+s)+2}, \quad \gamma = \frac{s}{m+s+3}, \quad h = \max \left\{ (L_n/n)^{1/[2(m+s)+2]}, \delta^{1/(m+s+3)} \right\}.$$

For $h \leq h_0$ and sufficiently small combined error, with probability at least $1 - \zeta$,

$$\|F^{\hat{T}} - F^T\|_{C^0([-R, R])} \leq C \left\{ e^{-\omega N} + (L_n/n)^\alpha + \delta^\gamma \right\}. \quad (\text{A.7})$$

No candidate-closeness assumption, analytic prior, supplied density jets, or inversion on arbitrary density tuples is used.

Proof. On the event of Lemma A.1, denote its right-hand bound by e . The true limiting tuple is within $e + C_0 e^{-\omega N}$ of the estimates, by (A.2). Density of the countable image makes this also an upper bound for the infimum in (A.6). The triangle inequality gives

$$\|f^{\hat{T}} - f^T\|_{C^m(Q)} \leq 2e + 2C_0 e^{-\omega N} + n^{-1}.$$

Both sides of this comparison are realizable limiting tuples. Apply (A.4) when this bound is small. In particular no factorization of the empirical or finite-flight density is asserted.

To simplify the rate, write $h_n = (L_n/n)^{1/[2(m+s)+2]}$. At $h = h_n$ the square-root term equals h_n^s ; the linear term is $h_n^{m+2s} \leq h_n^s$. Both stochastic terms decrease when h increases. Also $h^s \leq h_n^s + \delta^\gamma$ and $\delta h^{-m-3} \leq \delta^\gamma$, with the zero-noise term interpreted as zero. For large enough n , $n^{-1} \leq h_n^s$ and $h \geq n^{-1}$. These facts prove (A.7).

For repeated independent preparations, each observation is a failure or a successful mark. Its law is the mixture of a failure atom and $p_{N,b,d}^T f_{N,b,d}^T$. It can therefore be generated by an independent Bernoulli success sequence and an independent sequence of marks with that conditional density. The first n successful marks are independent with the required density. No mixing assumption about a single unreset billiard trajectory is being substituted for independent preparations. $\square$

A.4. A deterministic budget with failures charged.

Theorem A.3 (Charged smooth recovery). *Perform exactly B independent full-phase preparations in each of the J groups, all at $N \geq N_0$ flights. Define*

$$n_B = \lfloor Bp_*(N)/2 \rfloor.$$

Assume $n_B \geq 2$ and $Bp_(N) \geq 8 \log(J/\zeta)$. Use the first n_B successes in every group and the preceding estimator; return any fixed member of $\mathcal{C}$ if a group has fewer successes. Then (A.7) holds with $n = n_B$ and probability at least $1 - 2\zeta$, subject to the same small-error and bandwidth conditions. Every one of the JB preparations is charged.*

In particular, put $L_B = \log(C_2 JB/\zeta)$ and, for large enough B , choose the returning flight number

$$N_B = P \left\lfloor \frac{\alpha \log(B/L_B)}{P(\omega + \alpha\Gamma)} \right\rfloor \geq N_0. \quad (\text{A.8})$$

There is a constant C such that

$$\mathbb{P}_T \left(\|F^{\hat{T}} - F^T\|_{C^0([-R, R])} > C \{ (L_B/B)^\beta + \delta^\gamma \} \right) \leq 2\zeta, \quad \beta = \frac{\alpha\omega}{\omega + \alpha\Gamma} > 0. \quad (\text{A.9})$$

The assertion is uniform over the stipulated marked class. Constants, orders and exponents need not be uniform in a varying period or a vanishing hyperbolicity margin.

Proof. The number of successes in one group is binomial with mean $\mu = Bp_{N,b,d}^T \geq Bp_*(N)$. The elementary lower-tail bound $\mathbb{P}(M < \mu/2) \leq e^{-\mu/8}$ implies

$$\mathbb{P}(M < n_B) \leq e^{-Bp_*(N)/8}.$$

For example exponential Markov inequality at parameter $\log 2$ gives $\exp\{-\mu(1 - \log 2)/2\}$, which is no larger than $e^{-\mu/8}$. The union bound gives probability at most ζ of any insufficient group. By the mixture representation at the end of the preceding proof, the first n_B marks have the

requisite product law independently of the success indicators. Intersecting the sufficient-success event with the density-estimation event loses at most another ζ . Arbitrary bounded recording errors are still handled pathwise.

Write $N_* = \alpha \log(B/L_B)/(\omega + \alpha\Gamma)$. Then $N_B \in (N_* - P, N_*]$, so

$$e^{-\omega N_B} \leq e^{\omega P} (L_B/B)^\beta, \quad n_B \geq cB^{\omega/(\omega+\alpha\Gamma)} L_B^{\alpha\Gamma/(\omega+\alpha\Gamma)}$$

for sufficiently large B , with the floor absorbed into $c > 0$. The logarithm L_{n_B} is bounded by CL_B since $n_B \leq B$. Substitution gives $(L_{n_B}/n_B)^\alpha \leq C(L_B/B)^\beta$, proving (A.9). The sufficient-success condition follows for fixed confidence and large B ; for varying confidence it remains an explicit requirement. The bandwidth and small-error restrictions also remain explicit, especially when δ does not tend to zero. $\square$

The factor Γ in (A.9) records the cost of rarity. It would be absent if successful bridges were incorrectly treated as free observations. With a bounded period length, the total physical observation time is at most $CJB(1 + N_B)$, hence $O(JB \log B)$ at fixed confidence. The supplied polygon and frames and the two exact offsets are still required. This is a finite-sample result for bounded endpoint recording errors, not arbitrary contamination, uncalibrated clocks, a single correlated trajectory, or a claim of minimax optimality.

The estimate controls complete profiles and hence applies to the actual flat family of Proposition 5.6, when restricted to a small compact parameter interval. To make this consequence precise, fix $0 < |u_*| < R$ in the region where its cutoff is one. The two tables at parameters 0 and $\varepsilon > 0$ have profile separation at least $\Delta = \varepsilon e^{-1/u_*^2}$. Whenever $C\{(L_B/B)^\beta + \delta^\gamma\} < \Delta/3$, (A.9) and comparison of the estimated profile's distance to these two candidates give a test whose error at either table is at most 2ζ . The remote area correction does not change this contact value. Thus flat information is accessible to a finite charged experiment at every fixed nonzero scale, although its cost can be very large as u_* approaches zero. Equal Taylor jets alone still supply no such test. No assertion about equal full marked length spectra, unvisited smooth arcs, or normalized density jets is introduced by this observation.

REFERENCES

- [1] J. De Simoi, V. Kaloshin and M. Leguil, Marked length spectral determination of analytic chaotic billiards with axial symmetries, *Invent. Math.* **233** (2023), 829–901. doi:10.1007/s00222-023-01191-8.
- [2] D. Finamore and M. Leguil, *A CAT(0)-approach to the marked length spectral rigidity of Sinai billiards*, arXiv:2510.18983v1, October 21, 2025. <https://arxiv.org/abs/2510.18983v1>.
- [3] *Independent referee report on A2, revision 65*, author-requested AI-assisted memorandum, September 16, 2026, Section R65-M5. Repository [TrillionniumFoundation/theta-theory](#), frozen memorandum. Cited for the Riccati derivative comparison, not for a global inverse or a commissioned journal decision.
- [4] E. Giné and A. Guillaou, Rates of strong uniform consistency for multivariate kernel density estimators, *Ann. Inst. H. Poincaré Probab. Statist.* **38** (2002), 907–921. doi:10.1016/S0246-0203(02)01128-7.
- [5] L. Noakes and L. Stoyanov, *Lens rigidity in scattering by unions of strictly convex bodies in $\mathbb{R}^2$* , *SIAM J. Math. Anal.* **52** (2020), 471–480. doi:10.1137/19M1270409. Corrected planar theorem: arXiv:1803.02542v2, Theorem 1.1.
- [6] *Independent referee report on A2, revision 72*, author-requested AI-assisted memorandum, September 17, 2026, Section R72-M5 and Section 7. Repository [TrillionniumFoundation/theta-theory](#), frozen memorandum. Cited for the finite-flight clock identity and the request to display the weighted-order condition; not a commissioned journal report or a proof certificate.
- [7] K. Hamm and L. Huang, *CUR Decompositions, Approximations, and Perturbations*, arXiv:1903.09698v2, 2019. arXiv:1903.09698v2.
- [8] *Independent referee report on A2, revision 74*, author-requested AI-assisted memorandum, September 17, 2026, Section R74-M2. Repository [TrillionniumFoundation/theta-theory](#), frozen memorandum. Cited for the affine gate-conditioning calculation, not a commissioned journal decision or a mathematical certificate.